Perceptible Capture: A Speculative Design Study of Public Privacy for People with Visual Impairments

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Abstract

Cameras carried by other people increasingly shape public space. For blind and visually impaired bystanders, a phone or a pair of smart glasses gives no perceptible sign that it is recording, so the information needed to judge whether a recording is acceptable is unavailable. Prior work has examined the privacy risks of assistive tools themselves; the experience of visually impaired people as bystanders to other people's devices has received less attention. This thesis asks how they experience privacy in public spaces shaped by camera-based technologies, and how design might make those experiences more equitable. In this exploratory, qualitative study using speculative design, interviews with three visually impaired adults, questionnaires from two more and from a sighted guide, and three public online discussions were analysed thematically. The themes informed two speculative concepts, one built as a prototype and explored in a Wizard of Oz study with six sighted participants.

Participants judged recording by the recorder's intent and by being asked first, yet could rarely obtain the information that judgement requires, so their reported lack of concern follows from undetectability rather than indifference. None proposed restricting cameras; they asked for capture to be made perceptible. The sighted participants drew the same line between incidental and deliberate filming: the norm is shared, the means of telling when it has been broken is not. The prototype supplied awareness in almost every staged situation, yet each participant checked the alert by looking before acting, which a blind wearer cannot do. Verification therefore cannot be engineered away, and the question becomes where reliance sits. Here it sat with sighted companions, never discussed or arranged; a device moves it to a manufacturer, a capture signal to whoever regulates recording equipment. None removes the need for trust. A more equitable public privacy would begin with accountable trust and with a responsibility carried by those who record rather than by those who cannot see.

Keywords: visual impairment; bystander privacy; accessibility; camera-based technology; speculative design; Wizard of Oz; contextual integrity

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Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
ATM	Automated Teller Machine
ESP32	Espressif 32-bit microcontroller
LED	Light-Emitting Diode
MQTT	Message Queuing Telemetry Transport
PIN	Personal Identification Number
RP	Retinitis Pigmentosa
ROP	Retinopathy of Prematurity
SU	Stockholm University
VI	Visual Impairment / Visually Impaired
WoZ	Wizard of Oz

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1

Introduction

Digital technologies shape public space more than ever, changing how people find information, communicate and move through daily routines. For people with visual impairments these tools can widen access, but they also change how privacy is experienced in public. Privacy is not a rigid concept. It is a situated judgement that depends on local norms about what information is appropriate to share and how it moves between people [37], and preferences shift with context rather than holding steady across situations [1].

That judgement is harder to make without sight. Limited access to visual cues makes it difficult to monitor the surrounding environment, to know whether someone is eavesdropping, reading over a shoulder, or exposing personal details. People adapt with strategies that work: avoiding certain places, moving away from crowds, speaking quietly, or relying on trusted friends and family [3, 2]. Everyday tasks such as using an ATM, riding the bus or navigating shared areas become stressful when situational awareness is missing, and the strategies that manage that uncertainty come at a cost to independence and to control over personal data [2].

Among the technologies now common in public space, cameras operated by other people matter most for this thesis. Smartphones, wearable cameras, smart glasses and livestreaming equipment are held or worn by bystanders rather than run by institutions. A sighted person usually notices when a phone goes up, because posture and hand position signal it, and

can step out of the frame, change route, or avoid eye contact [44]. Someone who cannot see that has to infer recording from indirect sound, a shutter or an overheard conversation, and those sounds are easily missed [51]. The people least able to notice recording are therefore also absent from the informal conventions that grew up around visible cameras.

Existing research approaches privacy and visual impairment from two directions. One body of work examines the assistive tools themselves, showing that camera-based aids improve access and independence while introducing concerns about image sharing, misrepresentation, fairness and data handling [6, 7, 45, 50], and that users of visual assistance services expect transparency, safe storage and meaningful safeguards [8, 45, 50]. A second body examines the people on the other side of the camera, finding that bystanders will share information with visually impaired users but only within limits of what feels fair and appropriate [4, 6, 7]. Far less attention has gone to the position visually impaired people occupy most often in public: not as users of an assistive camera, but as bystanders to other people's devices, where the concern is missing privacy warnings and an unequal awareness of nearby recording [51]. The social dimension is largely absent too. Sighted guides, personal assistants and companions are present in many of these situations, and how they shape privacy practice is not well understood.

The problem underlying this thesis is that camera-based technologies have become difficult to perceive, and the people least able to perceive them are given no alternative means of knowing when they are being recorded. This is treated here as a question of accessibility rather than only of privacy. Two research questions follow. First, how do people with visual impairments experience public privacy when exposed to camera-based technology? (RQ1) Second, how might design interventions support more equitable public privacy experiences for visually impaired people? (RQ2)

The thesis answers these in two phases. The first combines a narrative literature review, semi-structured interviews, online questionnaires and public online community discussions, analysed thematically [17] to identify recurring tensions around inaccessible environmental information, public recording, consent and unequal awareness. The second translates those findings into speculative design concepts [11] that give form to the privacy concerns, adaptive behaviours and infrastructural inequalities the empirical work surfaced. Rather than proposing deployable solutions, the concepts function as critical provocations, examining how future recording

and assistive technologies might redistribute awareness, agency and responsibility in public environments. One concept was developed into an experiential prototype and explored with participants in a Wizard of Oz session [35, 22], focusing on how the interaction is understood rather than on technical performance, while the other remains conceptual. Throughout, public privacy is approached as a socially negotiated and context-dependent problem whose boundaries cannot be resolved through a single universal solution [40].

This thesis contributes to human–computer interaction research on accessibility and usable privacy in three ways. First, it gives an empirical account of how visually impaired people manage privacy in public spaces where recording devices leave no perceptible trace, including the finding that composure about being recorded is better read as an inference drawn from undetectability than as indifference. Second, it reframes public privacy as an accessibility issue, and argues that frameworks built on contextual judgement assume a perceptual access to context that not everyone has. Third, it presents two speculative design concepts and an evaluation of one of them, which together surface the tensions a wearable awareness device cannot resolve, chiefly that its alerts cannot be verified without sight, and point instead towards recording signals everyone can perceive.

2

Extended Background

This chapter conceptualises privacy as a matter of contextual norms rather than secrecy, notes the legal framework within which recording in public takes place, and reviews prior work on privacy and visual impairment: the everyday concerns of people with visual impairments (VI), the tensions introduced by camera-based assistive technologies, and the reactions of sighted bystanders to them. It closes with the case in which VI people are themselves the subjects of camera capture, which is where the gap addressed by this thesis lies.

2.1 Conceptualising Privacy

This thesis understands privacy as the preservation of appropriate information flows within social contexts rather than as the concealment of information. Following Nissenbaum, a violation follows from a breach of contextual norms rather than from the sensitivity of the information itself: health details belong within medical norms, and the same details reaching an employer do not [37]. These norms do not lapse at the entrance to public space, where declining to give one's name to a stranger remains reasonable [37], and the same logic holds online [1].

A second tradition treats privacy as boundary regulation. Altman describes it as the selective control of access to the self, a dialectical process in which a person opens and closes to others as circumstances change [9]. The two accounts are complementary here, one explaining why a flow of information is objectionable and the other why the same disclosure can be

acceptable when chosen and objectionable when compelled. Both rest on the same precondition: applying either requires knowing the setting and the audience, and that knowledge becomes less available when the surrounding environment cannot be perceived. It is this precondition, rather than the norms themselves, that the present work examines.

2.2 Recording and Accessibility in Law

No single rule decides whether a bystander may record another person in public. The GDPR does not apply to personal data processed by a person "in the course of a purely personal or household activity", an exemption limited to activities with "no connection to a professional or commercial activity" [27, Art. 2(2)(c) and Recital 18]. The Court of Justice reads it narrowly, and in Ryneš, decided under the earlier Data Protection Directive, it held that a home camera which also filmed a public footpath fell outside it [20]. Beyond data protection, recording in public is governed by national law, which differs between countries. Consent is therefore neither always required nor always optional, and being asked first is in most situations a social norm rather than a legal right.

Accessibility law treats the same spaces differently. The European Accessibility Act covers payment terminals and self-service terminals such as cash machines, ticketing machines and check-in machines "placed on the market after 28 June 2025", but Member States may let older terminals stay in service "until the end of their economically useful life", for no longer than twenty years after they entered into use [26, Arts. 2(1) and 32(2)]. Inaccessible interfaces of the kind participants describe can therefore lawfully remain in place into the 2040s.

2.3 Related Work

2.3.1 Privacy Concerns of People with Visual Impairments

Two factors shape privacy risks for VI people: a limited ability to monitor the surrounding environment, and a recurring dependence on others during everyday tasks [46, 14]. Ahmed et al. report that participants relied on helpers, often strangers, to handle documents, forms, and financial or medical information, which increased the risk of unintended disclosure aggravated by uncertainty over who else was within range. Assistive tools introduced vulnerabilities of their own: auditory output invited aural

eavesdropping, while magnified text widened the opportunity for shoulder surfing [3]. In a later urban study, these concerns proved inseparable from physical safety, which dominated participants' accounts of streets, transit and ATM booths, and lacking direct technological control, participants fell back on behavioural adaptations: avoiding certain spaces, relocating, restricting help to acquaintances, or accepting the risk outright [2, 16].

These strategies do not unfold in isolation, since moving through public space is itself a shared activity. Vincenzi et al. show that navigation is jointly constructed with sighted guides, with environmental awareness distributed across the pair rather than held by either partner alone [47], so what a VI person knows about their surroundings depends in part on what the companion notices and chooses to communicate. Hayes et al. identify a comparable cooperative dimension in privacy management itself: participants selected helpers strategically, with privacy an explicit criterion, yet the arrangement also introduced unintended disclosure and occasional friction over autonomy [31]. Privacy management therefore depends on interactions with others rather than on individual control alone.

2.3.2 Assistive Technologies and Privacy Implications for People with Visual Impairments

Digital assistive technologies for navigating public space have concentrated on the practicalities of mobility, such as establishing orientation and avoiding obstacles [18, 21, 19]. Privacy concerns intensify as such systems come to rely on cameras, since capture becomes a condition of their operation, bringing unintended disclosure and uncertainty about how the data is handled afterwards [8, 45]. Obfuscation, the standard technical remedy, fits this setting poorly. Participants resisted automated concealment because it removed their agency, and their needs were situational rather than fixed, so the same content might need to remain visible in one situation and hidden in another [8, 50, 23]. Blind participants also contended with algorithmic errors in detecting private content, and could not judge whether concealment had succeeded without seeing the result [50]. They wanted private content never captured, real-time information about who handles their data and why, and a role for blind people in designing privacy-preserving systems [45].

Each of these studies, however, examines cameras that VI people operate themselves, a position that preserves at least an initial measure of control over whether capture happens at all. This thesis concerns cameras operated by other people, or embedded in ambient systems, in public

spaces that VI people neither selected nor control. There, even that initial measure disappears.

A related body of work asks who is positioned to define the problems and the solutions in this space. Disability studies scholarship has argued that disabled people are routinely enrolled as sources of data about their own difficulties rather than as designers or critics of the systems proposed in response [36], and participatory and disability-led approaches answer this by treating disabled people as authors of assistive technologies rather than as end users consulted once the design decisions have been settled [12, 30]. The point bears on the present work, since what counts as a privacy problem, and what an acceptable response looks like, is often decided before any VI participant is involved.

2.3.3 Bystander Privacy Perceptions in Assistive Technology Contexts

Bystanders respond to cameras differently depending on how visible the device is. In an in-situ study, passersby adjusted their behaviour around a conspicuous smartphone camera but showed no reaction to a wearable one, while reporting afterwards greater unease about the wearable, since a device whose recording state cannot be read left them feeling more exposed. Most would have preferred to be asked, while conceding that consent from everyone in a public space is unworkable [44]. The remedy the authors propose, a visible indicator of camera activity, is itself inaccessible to VI bystanders.

Bystanders are more accepting of camera use when the wearer is visually impaired. This acceptance is limited to information a sighted person could ordinarily perceive, a limit framed through equality and equity, and comes with an expectation of knowing when an assistive device is collecting information at all [4, 16]. VI users and bystanders share an unease about automated inference, worrying about algorithmic bias and about the misrepresentation of traits such as ethnicity, weight or emotional state; an asymmetry compounds this, since the behavioural cues VI participants found most useful were exactly those bystanders were least willing to share [7]. This literature examines VI people as operators of camera-based technologies. The reverse position, in which they are the ones being captured, has received comparatively limited attention.

2.3.4 People with Visual Impairments as Bystanders around Camera-Based Technologies

Zhao et al. address this position directly and establish the problem of information asymmetry. VI participants were largely unable to detect ambient recording devices, whether smartphones, security cameras, or smartglasses that pass for ordinary eyewear. Left dependent on shutter sounds that operators can mute and on verbal notification of uncertain reliability, they could neither assess the risk of being recorded nor verify whether a request to stop had been honoured. Several described anxious resignation or a deliberate refusal to dwell on capture, and the authors suggest that low privacy expectations may reflect adaptation to conditions in which the perceptual and social mechanisms sighted people rely on are far less available. A double disadvantage follows: the audible output of participants' own assistive tools disclosed their activity to nearby sighted people, while the cameras trained on them remained undetectable [51].

This reorients the question from what VI people can access through their own devices to what it means to move through ambient camera capture without the means to perceive it. How visually impaired people navigate that exposure, and what role sighted companions play in mediating it, is the underexplored dimension this thesis takes up.

3

Method

This is an exploratory, qualitative and interpretive study. Privacy is treated as something people reason about and act on in specific situations, not as a quantity to be measured, so the study aims at detailed accounts of lived experience rather than distributions. The research strategy is speculative design, in which designed artefacts give a possible future a concrete form so that people can react to it, raising questions rather than solving a defined problem [25, 11]. This suits RQ2, which asks how design might support more equitable public privacy rather than which solution works best.

The study was carried out in two phases. The *formative phase* addresses RQ1 through a literature review, interviews, an online questionnaire and online community discussions, analysed thematically [17]. The *generative phase* addresses RQ2 by turning these findings into two speculative concepts [11], one of which was built as a prototype and explored with participants in a Wizard of Oz session [22].

3.1 Alternatives Considered

A survey was initially considered, since it could have reached more visually impaired people and shown how widespread concerns about cameras are. It was rejected because the thesis emphasises the reasoning of those unable to perceive recordings, and closed-response items fix the categories of an answer in advance and allow no follow-up questions [39]. A previous

study [51] adopted in-depth interviews to understand visually impaired bystanders' concerns, prompting the use of semi-structured interviews in this research for detailed personal accounts and follow-up questions.

Participatory design involving visually impaired co-designers was considered a stronger option for the generative phase and ethically preferable, since it would have let participants decide what counts as a privacy problem and what an acceptable response looks like. Scholarship in disability studies advocates for the involvement of disabled individuals in research [36], positioning them as designers instead of mere subjects [30]. It was rejected due to feasibility concerns. Co-design depends on repeated sessions with the same people, and the recruitment process for the formative phase resulted in three interviews. Asking these three participants for a series of workshops, with design materials made accessible without sight, would have placed a heavy burden on a small group. As a result, the concepts in Chapter 5 were derived from participants' accounts rather than co-authored with them, a limitation addressed in Section 7.7.

Two other design approaches were considered. User-centred design would have refined the alert around users' needs and tasks [38], but it assumes that the answer is a product for a defined user, whereas this study asks whether a product is the right response at all. Research through design would have produced knowledge by iterating on the prototype itself [52], but at this stage the prototype could only be tested with sighted participants, so repeated refinement would have optimised it for the wrong users. Speculative design instead keeps the prototype as a question.

3.2 Positionality

The researcher, being sighted and lacking personal experience with the studied phenomenon, influences the work by potentially misinterpreting participants' reactions and designing solutions that reflect a sighted perspective, rather than understanding the unique challenges faced by those who are not sighted.

3.3 Formative Phase

3.3.1 Literature Review

A narrative literature review was conducted before collecting empirical data, focusing on the limited research surrounding camera-based privacy exposure for visually impaired bystanders. The review examined four related areas: privacy and visual impairment, assistive technologies, bystander privacy, and visually impaired individuals as bystanders. Thirteen relevant papers were identified through searches in academic databases like Google Scholar, ACM, and IEEE Xplore, forming the basis of the related work section 2.3.

3.3.2 Semi-Structured Interviews

Three adults with visual impairments were recruited via a Stockholm University research project. Participants had to be over 18 and self-identify as blind or visually impaired; no restrictions were set on cause or severity, yielding a sample from congenital blindness to progressive loss. Semi-structured, in-person interviews (45–60 minutes) were conducted in English, with Swedish support if needed. Topics included privacy in public spaces, camera-based technologies, sighted guides, and future technological change (Appendix B). With consent (Appendix A), interviews were audio-recorded, securely stored on Stockholm University servers, de-identified, and transcribed with the AI transcription tool AudioConvert[1]. The transcripts were then checked manually against the recordings.

A separate guide was prepared for sighted guides/companions (Appendix B), but no companion interviews were conducted.

3.3.3 Online Questionnaire

An online questionnaire with open-ended and multiple-choice questions was shared on Reddit and in two Swedish Facebook groups for visually impaired people. It received three responses (two visually impaired adults and one sighted guide). Due to the small, exploratory sample, responses were treated as qualitative data and analysed with the interviews.

[1] `https://audioconvert.ai`. The provider states that uploads are deleted within 24 hours and not used for AI training.

3.3.4 Online Community Discussions

To capture unsolicited perspectives, public discussions on r/Blind and three Facebook groups were queried using 41 keywords (Appendix C) regarding privacy, cameras, and visual impairment. From ten screened posts, two Reddit threads and one Facebook thread specifically addressing public camera exposure were included. These threads were coded and analysed in the same process as the interviews and questionnaire responses, and contributed to the same three themes. Because contributors' visual impairments could not be independently verified, the material is treated as widening the range of documented accounts rather than as evidence of how common a view is, and Chapter 4 marks these sources as contributors rather than participants throughout.

3.4 Thematic Analysis

The interviews, questionnaire, and online discussions were analysed together using thematic analysis, a flexible method for identifying and interpreting patterns across qualitative data, following the six phases described by Braun and Clarke [17]. First, familiarisation began as each interview transcript was checked against its recording and continued with reading all the material before coding. Next, the material was imported into Taguette, where relevant passages were given short descriptive initial codes. This produced 54 analytic codes and 13 context codes, and the context codes, which record each source's vision and assistive tools, describe the sample rather than findings. The codes were then transferred to Excel to search for themes, grouping related codes into categories and the categories into candidate themes. These candidate themes were reviewed iteratively against the coded data and refined. Each theme was then defined and named, giving three final themes with nine categories (Appendix D). Finally, the themes were written up in Chapter 4, with quotations marked by source and disagreements between participants reported rather than resolved.

3.5 Generative Phase

Both speculative concepts come from the formative findings. The first responds to the request for an alert for the user (Section 4.3). The second responds to the expectation of being asked first (Section 4.2) by having cameras announce themselves to nearby devices and request consent, so that the burden moves from the person being recorded to the person recording.

3.5.1 Speculative Prototyping

One concept was selected for prototyping because the alert was the feature participants asked for most often, and because it is the concept whose central assumption can be examined by putting it in front of someone. Following Speculative Design principles, the prototype was not built as a technically complete system, but as a plausible interaction to explore awareness, agency, and technological mediation [11].

3.5.2 Wizard of Oz Session

The concept was explored in a Wizard of Oz session to see how participants interpreted the interaction and what questions it raised, rather than to test technical performance. The researcher manually controlled the prototype while participants perceived it as autonomous [35, 22].

Participants

Six sighted adults unfamiliar with the concept were recruited through convenience sampling from the researcher's personal network. No demographic data beyond gender were collected. Sighted participants were used because recruiting visually impaired participants was difficult, and at this exploratory stage the aim was to see how the interaction was understood. This also avoided exposing visually impaired participants to potentially distressing staged covert-recording scenarios at an early exploratory stage. Blindfold simulation was rejected because temporary visual deprivation does not reproduce the lived experience or navigational skills of visually impaired people [13], and because disability simulations can lead sighted people to judge disabled people as less capable [43].

Setting and Procedure

Conducted in a laboratory at Stockholm University, the 30–45 minute sessions involved two people: the researcher controlling the prototype remotely and an assistant acting as a stranger in staged scenarios. Contextual alerts were generated in real time and delivered via synthesised voice. Following the tasks, a post-session semi-structured interview (Appendix G) captured participants' immediate interpretations. Participants were subsequently debriefed on the manual operation and offered the chance to withdraw their data. The prototype, the control dashboard and the session protocol are described in Chapter 6.

Data and Analysis

Data collected included task videos (capturing physical responses [33, 32]), observation logs, and interview recordings. The post-session interview transcripts served as the primary dataset for a second, independent thematic analysis [17], yielding four themes (Appendix I). Videos and logs provided contextual support but were not thematically coded.

3.6 Ethical Considerations

The study followed established principles of informed consent, voluntary participation and confidentiality [39]. All active participants gave informed consent (Appendices A and F) and could withdraw at any point. The online questionnaire was run in Stockholm University's survey tool, which was checked for compatibility with screen readers. Identifying information was removed before transcription, all recordings and session materials were stored on Stockholm University servers, and participants are referred to by code throughout.

The online threads come from r/Blind and a public Facebook support group for people with retinitis pigmentosa, both of which anyone can read without joining. Users often perceive such spaces as private even when they are publicly accessible, so although the posts were used without individual consent, contributors are identified only by source code, and usernames and thread titles are omitted. Exact quotations can be traced back to their authors through a text search, and transforming them can reduce this risk [28]. Posts from the Facebook group, where members discuss their own sight loss, are therefore paraphrased, and quotations from Reddit are kept brief.

The Wizard of Oz sessions involved two deceptions: participants were not told that the alerts were generated by hand, or that the second person in the room was part of the study team. Both were necessary for participants to respond to the device as a device. Participants were asked to play a fictional character and never to enter their own details, and a debriefing immediately after the post-session interview explained both deceptions and gave them the chance to withdraw their data.

4

Results

This chapter presents findings from interviews and online questionnaires with visually impaired adults (VIP1 to VIP5), a questionnaire completed by a sighted person who has guided a visually impaired family member since November 2025 (SP1), so that account reflects an early stage of guiding, and community discourse from a retinitis pigmentosa support group on Facebook (FB) and two r/Blind threads (RD1, RD2). Table 4.1 lists each source. Throughout, *participants* refers to VIP1–VIP5 and SP1, who consented to take part. *Contributors* and *commenters* refer to the authors of the Facebook and Reddit material, who were not recruited and whose posts are treated as naturally occurring public discourse.

4.1 How Privacy Was Understood

Privacy was understood differently by different people. VIP2 said so directly: "I think privacy is very different for all people, every people" **VIP2**. It also varied within a single account: VIP1 dismissed public recording in a crowd but said he would question it if he were alone, and VIP3 denied thinking about privacy at all "I don't think about privacy" **VIP3** while describing constant self-monitoring and a clear objection to being overheard.

4.1.1 Privacy That Does Not Apply in Public

For three participants (VIP1, VIP3, VIP4) privacy did not apply to public space at all, and VIP5 worried only rarely. This was a considered position rather than indifference: recording usually cannot be detected, and what

Table 4.1: Detailed presentation of data sources.

ID	Source / role	Vision & tools
VIP1	Interview — visually impaired adult	Fully blind from birth; white cane, Meta smart glasses, Be My Eyes, family/assistant as guides
VIP2	Interview — visually impaired adult	Progressive vision loss (RP-type, familial), night blindness first; white cane, guide dog, hearing, phone magnifier
VIP3	Interview — visually impaired adult	Fully blind (lost remaining sight ∼3 years ago, after ∼20 years of decline); guide dog, white cane, smart glasses + WhatsApp guiding by wife
VIP4	Online questionnaire — visually impaired adult	Legally blind (ROP, age-related decline); white cane, navigation apps, screen readers
VIP5	Online questionnaire — visually impaired adult	RP, ∼5% capacity, central vision only, reduced colour contrast; white cane, phone magnifier
SP1	Online questionnaire — sighted person / guide	Guides their mother (visually impaired since November 2025; vision fluctuates with lighting and time of day)
FB	Facebook post + comments (retinitis pigmentosa support group; 11 comments, 9 contributors)	Multiple visually impaired contributors; cane and guide-dog users
RD1	Reddit r/Blind (25 comments, 14 contributors)	Original poster and commenters who are blind or low-vision; predominantly white-cane users
RD2	Reddit r/Blind (46 comments, 19 contributors)	Braille learner with low vision; broader community

cannot be detected cannot be avoided. VIP1 connected this to not seeing, returning to the explanation twice: "It can be a part of me not seeing . . . then I feel I maybe don't really understand this private and public thing. I just become one in the crowd" **VIP1**. Asked how often they worried about appearing in a stranger's photo or video, VIP4 answered "Never" and VIP5 "Rarely". In the community data the same stance sounded like resignation: "there's simply nothing I can do except to accept this is a part of being blind" **RD2**.

Feeling anonymous lowered the sense of risk, but only while a crowd lasted. VIP1 did not care in a busy place "If I am in the place with very many people, I don't care about it" **VIP1** but alone he would ask: "I will take a question to people why they are [filming] and see . . . where this purpose is" **VIP1**. An RD2 commenter drew the same line, distinguishing photographs of entire crowds from those that isolate a single person.

This calm coexisted with a working assumption of being watched. Because a camera cannot be spotted, VIP3 assumed he always was: "I try to be in that position that it doesn't matter if they take a photo of me now" **VIP3**. He made the comparison himself, to "some famous people, they always act like they are recorded" **VIP3**. Undetectability became its own reason not to react: "If you take the camera and I walk, how should I know that you have a camera?" **VIP2**.

Detection was possible but unreliable, and in every recorded instance it was late, accidental or second-hand. A Facebook contributor registered a car's behaviour rather than its camera, alarmed that it seemed to change pace as they walked, and learned of the photograph only when their son said that a woman was taking their picture **FB**. Another contributor caught sight of a camera in a small patch of remaining vision. VIP5 reported acting once aware "If I realised a recording was going on and I was not willing to be part of it . . . I moved away" **VIP5** but did not say how the recording became apparent.

4.1.2 When Recording Becomes a Threat

Where being recorded felt threatening, the trigger was rarely the recording itself but the intent behind it. A reason judged good could make an unwelcome photo acceptable, and for VIP1 that meant "to inform the people how we with visual impairment live our life" **VIP1**. VIP2 gave the same justification for answering strangers who ask about her guide dog, treating public education as what legitimates being looked at, and turned the test into a

question she asks directly: “What is the purpose . . . what are you doing?” **VIP2**.

Purpose is harder to read than that test implies. The motives attributed to photographers in the data were neither good nor bad but curious: two RD2 commenters independently guessed the woman had “possibly . . . never seen someone reading Braille before, thought it was neat” **RD2**. The original poster rejected the reading, having already confronted her: “[I] stopped reading and angrily asked . . . why she was making photos of me” **RD2**. Curiosity fits neither side of VIP1’s and VIP2’s test, and neither party could establish which applied.

Closely tied to intent was the fear of being taken out of context, which is not hypothetical. One guide-dog user’s photo appeared on a university magazine cover beside an article about financial aid he was not receiving, and the contributor stressed that there had been no permission to take the photo and even less to publish it, all the more so as he was not on financial aid **FB**. The objection was to losing control of the story attached to the image, which implied that having a disability meant receiving financial assistance **FB**, and in one thread it became being turned into content: “I don’t want to be the poster boy of some ‘activist’ on TikTok or YouTube that looks at me as an easy way to boost their page” **RD1**.

Contributors also described hostility in public that had nothing to do with cameras. One reported it of their son, a cane user, who had had bottles thrown at him, had been honked at while crossing the street, and had had food thrown at him in the cafeteria **FB**. Where recording was involved, the white cane was named as the trigger: “I’m always worried that . . . someone is gonna shove their iPhone in my face . . . because I have my white cane” **RD1**. The cane therefore does two things at once: VIP1 described walking with it as what let him disappear into a crowd, while the RD1 commenter expects the same object to attract a phone.

Partial sight complicates any response. A Facebook contributor with a small remaining field caught sight of a camera and said nothing, because speaking up would reveal that they could see it, and they were too frightened to say anything for fear of being exposed as faking their impairment **FB**. This is a single account, but it names a cost that does not arise for participants with no residual vision.

Not all of the data reads as threatening. Several Facebook contributors described the public as curious or well meaning, one reporting no cruelty at all in years of cane use and finding that people usually just wanted to help or wondered why someone carried a white cane **FB**. An RD1 commenter

expected bystanders to intervene on the blind person's behalf: "many people would most likely come to your defense if someone did that" **RD1**. Across the data, accounts of threat cluster where the recording was covert and the reason unknown; where the photographer's purpose was stated or obvious, no contributor in this data described the recording as a threat.

4.1.3 Privacy as Controlled Disclosure

A second understanding of privacy centred on control over information, measured against the home. VIP2 located privacy "at home when I close the door" **VIP2**, and added that disclosure is selective even in conversation: "when I meet people, I talk, but there is lots of things I don't tell" **VIP2**. VIP1 defined a violation as someone crossing that boundary: "go into my house ... and go over a line ... You don't for example go in and steal a computer or start hitting someone or asking questions in someone's home" **VIP1**.

Against that baseline, the everyday problem was self-service terminals. VIP5 described the exposure precisely: "self-service kiosks or ticket machines. Others might then see what I am putting in, which could be an infringement on my privacy" **VIP5**. VIP2 met a sharper version, where the interface does not merely expose information but demands it. Flat touchscreens with no tactile keys left her unable to enter a card PIN, and staff offered to enter it for her. She refused: "Never. Never" **VIP2**. Where the design gives no private way to act, completing the task means handing a credential to a stranger.

VIP5 framed privacy as the right to decide when to reveal a disability: "Privacy for me is that I should have the right to self-disclose my disability, not that disclosure should be called upon by the public environment itself" **VIP5**. This was tied to independence and to getting the same access to environmental and service information as people without a visual impairment.

VIP3 described the opposite situation, in which blindness worked as a shield. Because he cannot see his own screen he leaves it dark, "when I use my phone, the screen is always black" **VIP3**, so nobody else can read it either: "I'm more shielded to other people than you are because people can see what you're doing" **VIP3**. This was a by-product of how he uses the device rather than a privacy measure. His concern ran through sound instead: using earphones in public was "a little problematic ... 'cause I don't want others to hear" **VIP3**. Where a sighted user's exposure is visual, his is auditory.

Participants disagreed on whether their own tools exposed them. VIP5 noted that using a phone as a magnifier "does show the surrounding environment that I am likely living with a visual impairment" **VIP5**, while VIP4, asked the same question, answered "No" **VIP4**. What made the exposure acceptable to VIP5 was that it was chosen: "doing so is my active choice and hence a way to enjoy the right of self disclosure" **VIP5**. The tool discloses in both cases; the difference VIP5 draws is between disclosing by decision and being disclosed by the environment.

VIP2's account shows the choice is not always free. She described friends who conceal their impairment, declining a cane because "I will look stupid" **VIP2**, and distanced herself from that view. She had nonetheless resisted the cane for years, and what ended the resistance was not a change of attitude but the cost of remaining unreadable to strangers. Her son's argument was that concealment made things worse: "You have to start to use it. Then they understand why you behave in a different way" **VIP2**. Non-disclosure preserved privacy in VIP5's sense, but left her behaviour to be explained as rudeness rather than as impairment.

4.2 Living with Recording That Cannot Be Seen

Participants held expectations about how other people should behave, acted on them when they noticed something, and otherwise relied on sighted companions to notice for them.

4.2.1 Asking First: A Norm without Enforcement

The expectation voiced most forcefully was that people should ask first. VIP2 returned to it throughout: "You have to ask me before . . . they have to ask" **VIP2**; "I can say yes or I can say no" **VIP2**. She described an incident at a pub she organises for her association, where a man filmed without asking. What made it serious was who was in the room: the venue is used by blind and low-vision people, and the man had more residual vision than most of those present "he could do many, many, like some photos or films without no one knows" **VIP2**. Her response was to set the rule out loud: "If you're recording when I'm here you have to ask me. Otherwise, put down these things" **VIP2**. The detection gap is therefore not only between blind and sighted people: it opens wherever one person can see what others cannot, including inside a space set aside for visually impaired people.

SP1 made the same demand on behalf of their mother, framing it as dignity and naming the detection problem in the same answer: “Digital technologies increase the risk because people can easily record or take photos without permission, often without others noticing” **SP1**.

Not everyone made that demand. VIP1 saw no reason to withhold permission where no harm was meant: “if people not will injure me or make a bad thing on me” **VIP1**, then he did not object. The community threads showed no consensus. Within RD2, commenters held that “asking for consent is mandatory” **RD2**, that consent “isn’t mandatory, at least in my state” **RD2**, and that a recording may be “legal sure but still rude” **RD2**. SP1 pointed to the same uncertainty in their own jurisdiction: “in Serbia, recording is not fully managed according to the law, still” **SP1**, and an RD1 commenter reported the opposite position as settled: “where I live it’s totally legal to film anyone in a public space” **RD1**. The norm therefore sits between people who treat it as obligatory and people who treat it as unenforceable. Underneath the legal difficulty lies a practical one: a rule that you should ask first can only be enforced by someone who can tell when it has been broken.

One distinction stands out. VIP2 objected strongly to being filmed by strangers, yet was comfortable with another blind person using the same camera hardware, saying of a friend’s camera glasses: “I think it’s wonderful” **VIP2**. She generalised the point to assistive technology as a whole. What mattered was not the device but why it was being used and who chose to use it.

4.2.2 Responding Once a Recording Is Noticed

Responses ranged from direct confrontation to no response at all. VIP2 described confrontation as her standard practice: “I face. I face, always” **VIP2**. RD1 commenters described the same impulse in stronger terms, from “I would probably call them weird and walk away” **RD1** to “I’d tell them where they can shove their phone” **RD1**. One described a quieter method: “I moved the hand I had holding my cane to where I was subtly flipping off the camera . . . The phone immediately went down” **RD1**. A gentler option was to move away, as VIP5 did. Others did not respond at all: “No, I don’t . . . I never confront” **VIP3**; “You can’t control other people and you can’t live life fearing the worst case scenario” **RD1**. One RD1 commenter noted that formal recourse was not realistic either: “suing can be costly upfront, and I’d rather not go through the process at all” **RD1**. VIP1 saw no reason to act, and VIP4 had “never thought about it” **VIP4**.

VIP2 described the same split among people she knows, treating it as temperament rather than correct practice: "we are different, everyone. Even us who's can't see so much, we are also different people" **VIP2**. She placed herself at one end "I am a confronter" **VIP2** and her friends at the other.

The disagreement extended to whether the concern was reasonable at all. On the Facebook thread one commenter dismissed the original post as paranoia, and two others firmly rejected that reading, insisting that the concern was not paranoia **FB**. On RD1 a commenter suggested the person describing the fear "consider talking to a professional about the anxiety" **RD1**. That person had already raised the same doubt about themselves: "The first time they did it I thought maybe I was being paranoid" **RD1**, and described the concern growing after their daughter was born: "The anxiety got worse when my spouse and I had our daughter, and I started to think about how it might affect her as well" **RD1**. Not being able to confirm a recording leaves the question of whether the worry is warranted open, and open to being answered by others.

4.2.3 Unspoken Protection from Sighted Companions

Where recordings came to light, it was usually a sighted companion who spotted them, and this rarely rested on an explicit agreement. SP1: "We have not explicitly discussed privacy in depth, but I naturally try to protect her and stay aware of the environment when we are outside" **SP1**. VIP2's son confronts photographers without ever having been asked to "if there's someone who would photo me, he said, 'Why do you do this? Have you asked us?' " **VIP2** and she was explicit that this was not arranged: "No, he just learned himself" **VIP2**. VIP3 said the same of his wife: "she's never said anything about that . . . I haven't instructed her" **VIP3**. VIP1 and VIP4 also reported that privacy had never been discussed with the person guiding them. VIP5 attributed this to a lack of concern rather than to a tacit arrangement "we never talk about it as I do not have any major concerns" **VIP5** though a guide was still described moving them out of a space "not to interrupt the others' feed" **VIP5**.

This arrangement depends on the companion being present and attentive, and in these accounts it fell to family members. When a companion was not there, technology stretched the same relationship across distance: both VIP1 and VIP3 streamed video to a relative for live guidance, VIP3 "through WhatsApp and my glasses with my wife" **VIP3** and VIP1 over the same messaging apps. The same camera and connection that bring in a remote helper also open a further route to being seen.

4.3 What Participants Asked For

Requests fell into three groups: an alert for the user, a recording signal perceptible to everyone, and accessible public infrastructure. None of the participants proposed restricting cameras. What they asked for was that visual information be made available in a form they could hear or otherwise sense. VIP1's distrust of existing tools indicates the accuracy such a tool would need: he had abandoned navigation apps as insufficiently reliable, "I don't dare use a phone to navigate" **VIP1**.

4.3.1 An Alert for the User

The tool discussed most often was a background alert that runs on its own, warns with a sound when a recording is happening, and stores nothing. VIP2 raised it herself: "on the glasses or something that boop, boop, boop ... 'now there's someone, it's recording' " **VIP2**. She wanted it so that she could then act: "So I know what is that sound and then I can ask the person, 'What are you doing?' " **VIP2**.

VIP1 and VIP3 endorsed the idea on qualified terms. VIP3 made his conditional: "If I would care, there should be a sound telling that somebody is taking photos ... But for me, it's not a big deal" **VIP3**. VIP1 described the mechanism concretely — an application "that sends you warnings that right there on your right side is a person filming around you" **VIP1** and specified that it should run unattended and keep nothing: "the tool must ... not save the material ... it is enough that the tool has an algorithm to warn" **VIP1**. The demand is therefore strongest in VIP2's account and qualified in the other two.

VIP1's constraint matters: the tool has to detect recording without recording, or it reproduces what it is meant to warn about.

4.3.2 A Recording Signal Everyone Can Perceive

VIP2 wanted the recording to be visible to everyone nearby, not only to her. Discussing camera glasses, she was explicit that the problem is not confined to people who cannot see: "I don't know. And you'd maybe don't know either ... You never know when they are filming you. Never" **VIP2**. She contrasted this with a raised phone, which at least announces itself, and wanted something equally obvious: "lights so the other one could see ... as ambulance or something, a little blinking" **VIP2**.

VIP2 drew the general conclusion herself, reaching for an example with no connection to blindness. She described teenage girls who avoid the school showers after gym for fear of being filmed: “So it’s for everyone, I think. Not just for us . . . But for us, it’s more than, more a problem, because we can’t see it” **VIP2**. Her account distinguishes two things. The recording signal has failed generally, so blind people are not missing a cue that sighted people reliably receive; what is specific to them is that they cannot catch it by chance, and so bear the failure completely rather than partly.

4.3.3 Accessibility as a Privacy Measure

VIP5 and SP1 looked to public infrastructure and treated accessibility and privacy as the same issue. Where kiosks, public spaces and digital services are unusable, or do not support smaller languages, people are forced to ask a stranger for help with something private, and disclose information in the asking. VIP5 made the link explicit: “signage is good, large, and clear enough even for me with low vision so that I would not have to ask strangers for help” **VIP5**. The requests were institutional: “self-service kiosks should follow the new accessibility directive . . . Public spaces, and urban environments in general, should be designed to follow accessibility standards (at the minimum), and aspire to consider Universal Design principles” **VIP5**. SP1 asked for the same in language terms: “Better voice assistants in smaller languages would also help them be more independent and less exposed” **SP1**.

VIP2 raised the language question from the user’s side. She had decided against smart glasses that several friends use because they operate in English, and expected to buy a pair once “they will be in Swedish” **VIP2**. A tool that does not speak the user’s language is, for her, not available at all.

For these participants, the most basic privacy measure is making public places accessible. Usable voice assistants and compliance with accessibility standards are described here not as conveniences but as part of privacy itself, because they remove the need to hand private information to a stranger.

5

Design

This chapter turns the findings from Chapter 4 into two speculative design concepts, each shown through its main features and through short fictional scenarios set in ordinary public situations. The scenarios are collected in Appendix E and pointed to from the feature that each one illustrates. The aim is not to offer a finished product, but to picture what public life might look like if the ability to notice recording, and the ability to act on it, were shared more equally between blind and sighted people. The features came out of the interviews, the online questionnaire, the public online discussions, and existing work on accessible privacy. Privacy is not presented as something that can be settled once and for all; the concepts propose directions to think with rather than solutions [40].

The first concept, ***360° Privacy-Aware Smart Glasses***, is a wearable system that converts visual cues about recording into feedback a blind user can perceive directly, making forms of observation noticeable that would otherwise pass unremarked. The second, ***Real-Time Privacy Mediation System***, is a signalling layer running between devices, which intervenes at the moment of capture so that recording becomes visible, and negotiable, while it is happening.

5.1 360° Privacy-Aware Smart Glasses

The first concept pairs AI-assisted smart glasses with a haptic bracelet worn at the wrist. The glasses monitor the space around the wearer for camera-based devices that people nearby are using to record, photograph, or livestream,

and for suspicious behaviour such as shoulder surfing or actions that could put the user at risk of harm. This monitoring is processed locally on the device, and the results are translated into directional audio through bone conduction, vibration at the wrist, and coloured light along the frame. The bracelet doubles as the control surface, used to permit a recording for a short time, adjust settings, or begin a forensic recording.

Subtle, private feedback. Nothing the system reports is spoken aloud. Information reaches the wearer along three channels that neither block the ears nor draw attention: sound conducted through the bone of the skull, vibration at the wrist, and coloured light on the frame. Hearing stays available for the street, and what the system says stays with the person it is speaking to. Both points follow from how visually impaired people describe managing in public, keeping headphones off so the surrounding sound remains available to them, and preferring alerts that do not make them conspicuous [3, 51]. Every feature that follows depends on this (Scenarios S1 and S2).

Biometric activation. Glasses are at risk of being separated from their owners, which opens the door for an unauthorised user to reach sensitive information about bystanders. The glasses could therefore read the biometrics of whoever puts them on and work only for their owner, closing a route by which the information they collect could reach someone it was never meant for [4, 2] (Scenario S1).

Trusted contacts. People the system counts as trusted are enrolled in advance through the service provider rather than added in the moment, since managing complex privacy settings through a limited wearable interface while navigating public space adds cognitive load the wearer does not need [51]. The list builds into the system something participants already relied on, since they described turning to familiar people for help with privacy. When the glasses recognise someone on it, they do not alert the user, filtering out the cameras of friends and family [4] (Scenario S1).

Outward-facing signalling. Coloured LEDs run along the frame and change with what the system currently registers: a trusted person nearby, a recording in progress, a situation escalating, an emergency. This channel is meant for everyone except the wearer. The other channels keep information with the user, while the lights send part of it back out, so that something known

only to the device becomes readable to the people around it. That matters because recording in public often goes unnoticed by those it captures [49, 7, 4] (Scenario S1).

Directional recording detection. The glasses look continuously for the signature of an active camera, and what reaches the wearer is a direction and a distance rather than a general statement that recording is taking place. That distinction carries the feature. A camera behind a blind pedestrian is not simply unseen, it falls outside every channel available to them, and giving the event a bearing restores some sense of what lies in front of and behind the body, keeping the space nearby readable as a personal "privacy bubble" [2]. An alert is of little use without the spatial context needed to respond to it, since knowing that a camera is present says nothing about where the person is or which way to move [3] (Scenario S2).

The concept goes further and alerts only when the wearer is inside the camera's frame, rather than reporting every recording nearby, which is what participants asked for [51].

Negotiated, temporary consent. The system does not decide on the user's behalf. It reports, and the decision is made at the bracelet, for that encounter only. Two findings pull in opposite directions here: people want to be asked before they are recorded, and they are uneasy about assistive systems that act for them without being asked. Granting permission one instance at a time is an attempt to hold both [8, 51, 44]. Because the permission runs out by itself, agreeing once neither creates a lasting arrangement with a stranger nor leaves the user having to confront anyone in order to end it (Scenario S2).

Processing locally. All sensing happens directly on the glasses. Camera frames never leave the device; they are analysed in the background and immediately discarded, with nothing sent to a remote server. This addresses two concerns from prior work: that sensitive data passing through the cloud might be stored or misused outside the user's control [31], and that a wearable camera captures everyone in view, whose privacy visually impaired users often treat as carefully as their own [51]. Images of people nearby exist only for the fraction of a second they are needed to trigger an alert.

Suspicious behaviour detection. The system attends to how nearby people are placed and how they are behaving, treating standing closer than the setting warrants, or holding attention on the user, as a sign of intrusion [2]. This covers visual eavesdropping, aural eavesdropping and shoulder surfing [3]. The alert arrives through bone conduction rather than aloud, because announcing it publicly would expose the wearer in the exact way the feature is designed to prevent [3, 51] (Scenario S3).

User-controlled forensic mode. Processing remains local with one exception. If an encounter turns threatening, the wearer can activate a forensic mode that preserves a recording and sends it off the device, streamed live to a trusted contact or sent securely to the authorities for a later report. Storing this evidence remotely matters because a recording kept only on the glasses would disappear if an assailant took the device [2]. The feature sits on a tension rather than settling it. Evidence is valuable during an unsafe encounter, but a system that records automatically would bring back the loss of control it was built to prevent [8], and sending footage elsewhere raises the question of who else might reach it [31]. The wearer is therefore the only trigger and decides who receives the footage (Scenario S3).

5.2 Real-Time Privacy Mediation System

The second concept is not limited to a single user; it works between devices. A recording device broadcasts a simple, machine-readable signal when active, a nearby person's device reads it and replies with their privacy settings, and the system can then allow the recording quietly, pause for consent, temporarily stop a livestream, or remove the person from the final shot. This is less a single product than a shared layer built into public recording, and it assumes common standards that do not exist yet. That assumption is intentional. SP1 described recording law in their country as unsettled, and the community discussions could not agree whether asking for consent in public is a legal requirement or a courtesy. Where the first concept helps one person perceive what is already happening, this one changes what is allowed to happen, moving the burden from the person being recorded to the person recording.

Device-to-device mediation. Negotiation is a standard setting on all devices, so that consent is requested rather than assumed [44, 51]. Every camera broadcasts its activity in a format other devices can read, and nearby

devices reply with their owner's preferences over a short-range link such as Bluetooth. Users choose one of three capture settings. Under *always*, recording continues without alerts. Under *ask every time*, the camera user waits for permission. Under *never*, consent is denied by default. Any setting can include a list of trusted people who bypass these rules. The exchange takes seconds, so the outcome depends on who is present and how well they know each other [2, 51] (Scenarios S4, S5 and S6).

Trust-based suppression. In familiar settings, friends and family own most of the nearby cameras, and constant alerts would make the system tiring to live with. Captures by trusted contacts therefore trigger no tones, vibrations or decisions, using the same trusted list as the first concept. This follows the argument that privacy systems should adapt to context and existing relationships rather than treating every camera the same [3, 51] (Scenario S4).

Automatic interruption. If a user selects *never*, the response depends on the media type. For photos or recorded video, the recorder's phone removes the person automatically. Live streams cannot be edited later, so the broadcast pauses and resumes only when the person leaves the frame. Here the system shifts from noticing cameras to intervening, and the limit falls on the person holding the camera rather than the person in front of it (Scenario S5).

Localised notifications. Each side receives information in the form that works for them. The person recording sees a message on screen in their chosen language, while the visually impaired person receives a short audio or haptic prompt, and only when the system needs their input. Handling language locally addresses a concern participants raised directly: when a system is inaccessible or in the wrong language, users must ask strangers for help, and disclose private information in the asking (Scenarios S5 and S6).

Removal over blurring. When a user denies consent, the system removes them from the frame rather than blurring their face. The distinction matters to users, who describe blurring as a gesture towards protection rather than protection itself, and who suspect it can be undone later [45] (Scenario S6).

Taken together, the two concepts treat privacy less as a rigid line set by law than as an ongoing agreement worked out between people, public

spaces, and the devices around them. They also carry unresolved tensions, some of which push against the personal accounts they are built on. Those tensions are examined in Chapter 7.

6

Wizard of Oz Session

Chapter 5 described two concepts on paper. This chapter takes the first off the page. One function of the 360° Privacy-Aware Smart Glasses was built as an experiential prototype and put in front of participants, so the concept could be met as an event rather than read as a description.

6.1 Design

Among the features considered, the alert system was developed to inform the wearer when a camera is aimed at them or when someone views their screen. Other functionalities like biometric activation, recognition of trusted individuals, and forensic mode were excluded from participant knowledge. A bracelet was also introduced, as the alert alone lacked interactivity; the device communicates, but the wearer cannot respond. A gesture, a tap to mute, was included but kept intentionally vague to observe participant assumptions.

Two scenarios were staged to test a system's response. In the first, a stranger films the participant, with the glasses providing instructions on how to step out of the frame. In the second, someone observes while the participant enters financial details, with the glasses only naming the event. The first scenario allows for a clear physical action, while the second lacks an actionable response, leading to an alert that simply informs the user of the situation.

6.2 Apparatus

In the setup, all alerts were manually generated by the researcher, as the rig itself detected nothing. The prototype's believability was enhanced by its physical weight, sound, and its resemblance to an actual product, leading participants to engage with a realistic device rather than an obvious prop.

The prototype used a pair of Meta Adventurer glasses, paired to the researcher's laptop as a Bluetooth speaker (Figure 6.1). Elastic straps run from the arms to a 3D-printed pod that sits centred behind the head. The pod holds soft padding, a dummy battery for weight, and a deep tunnel with a printed lens behind a clear shield, so that it reads as a rear-facing camera. The bracelet is worn on the non-dominant wrist. It is a printed case holding an ESP32 microcontroller, a battery and a vibration module. The module is mounted through the wall of the case, because vibration from inside it could not be felt well. A strip of conductive tape on the lid works as the tap sensor.

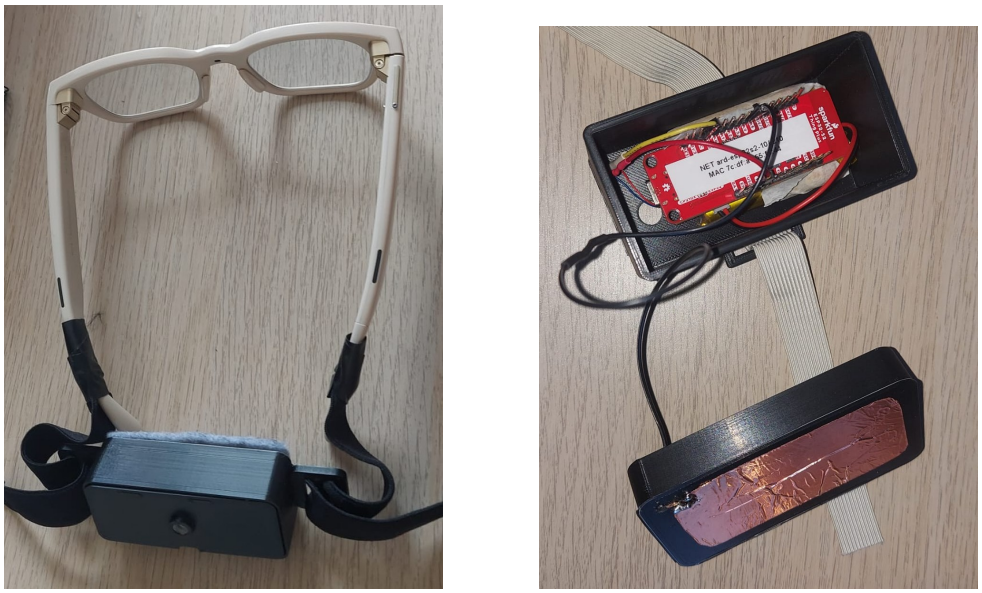


Figure 6.1: The prototype: glasses with the centred pod (left) and the bracelet (right).

The browser dashboard (Figure 6.2) manages the session by following the protocol, providing buttons for participant reactions that initiate follow-up lines and a countdown simultaneously. Alert texts are live-typed to match the actual position of the helper. It also serves as an observation log, timestamping every cue, spoken line, vibration, and detected tap, with records

exported post-session. Communication with the bracelet is via MQTT, facilitating bidirectional taps and silent phone notifications to the helper, ensuring precise timing without visible signals in the room.

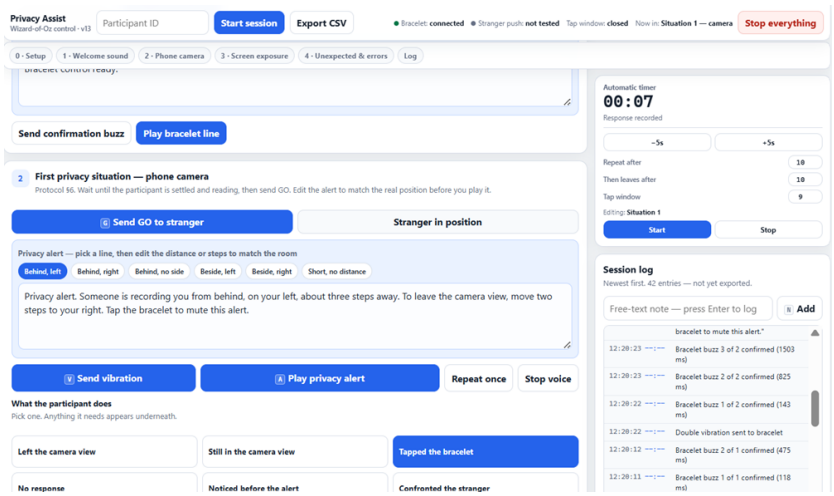


Figure 6.2: The control dashboard used to trigger alerts, cue the helper and log the session.

6.3 Procedure

After consent (Appendix F), participants were fitted with glasses and a bracelet to engage in a fictional scenario. They were instructed to play a character, avoiding personal details, and were unaware that alerts were manually controlled by a study team member. Following the fitting, participants interacted with an information packet while the helper pointed a phone at them, and the researcher triggered the alert from the dashboard. The device provided real-time feedback on their actions, including when they left the camera view. During a financial form completion, the helper approached, prompting another alert without further commentary. If a participant did not mute an alert, it was played a second time before the helper withdrew.

Participants were allowed to react freely during the protocol, with various actions such as moving, changing seats, or speaking to the helper considered valid. The interview took place immediately after, followed by a debrief. For full details, see Appendix H.

6.4 Results

The six sessions produced three kinds of material: the dashboard log, the task video and the transcribed interviews. How they were collected and analysed, and why sighted participants were used, are described in Chapter 3. Where a participant's account and their recorded behaviour disagree, both are reported below. Errors in the setup are listed in Appendix J; where a participant's response can be traced to one of them, it is left out of the matching claim.

No demographic data were collected beyond gender. All six participants were adults, five women and one man.

6.4.1 What participants did

Table 6.1 summarises the sessions. The reaction descriptions come from the video and the tap records from the log.

Table 6.1: Participant behaviour in the two situations.

ID	Situation 1, camera	Tap	Situation 2, screen	Tap
P1	Looked at the helper, moved but not as instructed, later came back	Yes	Brief look, tapped, carried on	Discarded
P2	Paused; looked at the helper without acting	No	Brief look, tapped, muted	Yes
P3	Looked at once, stopped reading, moved as instructed, tapped	n/a	Glanced sideways, muted, carried on	n/a
P4	Brief look, changed seat as instructed, tapped	Yes	Brief look, tapped, carried on	Yes
P5	Saw the phone go up before the alert, covered her face, moved as instructed, asked the helper why	Yes	Looked at the helper and covered the screen	No
P6	Ignored the alert, looked around, stood up and turned her back to the camera	No	No visible response; chose not to mute	Discarded

"Discarded" means contact was detected but fell outside the window in which the system would accept it. "n/a" means P3's log was lost; the video shows a tap in both of her situations.

Three patterns stand out.

The device noticed first. Only P5 experienced a situation before being informed, due to the phone appearing in her view, and the alert was delayed. Excluding the errors and two false alarms for P5 and P6, the prototype provided awareness in all other staged situations.

Muting followed a look, not a reflex. Participants looked at the helper before tapping, with each tap occurring several seconds after an alert. The log only records contacts within a predefined window after each alert, discarding any irrelevant early contacts. Consequently, the timings provide evidence for registered taps rather than the speed of initial contact. The video confirms that muting followed a decision rather than preceding one.

Movement was mixed. P3 and P4 followed the alert, while P6 chose her own direction. P2 did not move, and P1 received a contradictory alert, so her movement is not counted.

The responses varied between scenarios; in the camera situation, P3 and P4 stopped and moved as instructed, whereas in the screen situation, four participants looked briefly and resumed their tasks. All four tapped, but P1's tap fell outside the accepted window, so three alerts were muted. The camera situation occurred first, leading to habituation as a possible explanation, as noted by P3's comment on no longer perceiving a threat in the second situation.

6.4.2 Theme 1: Awareness Benefits and Costs

Increased Awareness The function participants liked best was the one the study was built around. Five of the six missed both situations until the device spoke, and several gave the credit to the device rather than to their own attention.

> "I was reading the text and I was not looking at… someone is recording. But this stuff that it send me, an alarm, and I can recognise that the person is recording." **P1**

P2 described the same order of events. P5 went further and treated the detection as something a person could not do at all, since a raised phone gives nothing away: "I would not, myself, be able to notice someone is recording me because people have their phones all the time and I don't know if they're looking at something, or if they're recording me" **P5**.

What they valued was being told in time to decide for themselves. None of the six said the device stopped anything from happening, and none asked that it should.

Situations of Use Asked where the device would help, participants described moments with a lot at stake, or moments when they cannot watch

their own surroundings: entering a PIN at a cash machine, opening BankID on the metro or the bus, confidential work, being alone somewhere unfamiliar at night, and watching children in a playground. P5 framed it as assistive rather than protective: “If someone needs assistance in their, like, everyday life, if they have, um, not so good sight or hearing or anything that could, like, make them not be fully aware of their surroundings, this could really be a good device” **P5**. P1 thought of a carer rather than the wearer.

Discomfort and Interruption The awareness was not always comfortable to receive. Two things caused discomfort. The first was not knowing what the device would do, and it passed after the first alert. P5 and P1 both described the opening moment this way, P5 saying that “The first time I got a bit, uh, shocked because I wasn’t ready for it to tell me that something is happening” **P5**. The second was the alert arriving in the middle of a task that already felt sensitive: “when I was trying to focus on something then I was kind of interrupt by the alert and also I was noticed that I was being watching by someone. It kind of make myself nervous” **P4**.

Participants read that discomfort in opposite ways. P3 treated it as useful, more or less as how the device works: “if you want to protect your stuff... you get that stressor from it” **P3**. P6 saw the same feeling as a possible harm, saying the device “makes people feel more worried about what’s happening, who are recording their behaviour” **P6**. Both readings come from the same design. That argues for letting the wearer control how loud the alert is, rather than deciding the question for them.

6.4.3 Theme 2: Alert Interpretation and Control

Warning vs. Instruction The camera alert ended with an instruction to move, and this changed how the whole message was heard. P3 followed it more fully than anyone, and had not understood it as a warning at the time:

> “At first, like, I just followed the instructions. I thought that was, ‘Okay, I have to follow this,’ but then realised, ‘Oh, it’s to warn me of something,’ and didn’t realise that until later.” **P3**

P4 described his first response in the same terms, as compliance rather than judgement. Doing what the alert says is therefore not the same as understanding it.

Understanding the System A related gap showed up at the other end of the session, in one participant’s account. P6 waited the whole time for

guidance that never came: “I was waiting for more guidance, for alerts again. I thought that after some minutes, for example, it will tell me go through the laptop or telling me something” **P6**. She had taken the activation line to mean that the device would talk her through the task, and she named this as the reason she felt confused. Neither the activation line nor the alerts say what the device is for.

Multimodal Feedback The two channels split the work between them. None of the participants said the vibration carried meaning of its own. It caught attention and handed over to the voice, which P5 described as a gap rather than a sequence: “It, it did vibrate, but I didn’t understand if it, like, it gave me some signs. It just... Like, it vibrated and then the glasses talked to me” **P5**. P2 thought the voice alone would have been enough. P4 made the case for keeping both, for reasons of attention rather than content: while typing he would “kind of ignore the sound” **P4** and miss the opening words, and the vibration was what told him an alert had arrived. A spoken alert should not put anything critical in its opening words.

Interaction Ambiguity The tap was the only thing participants could say back, and they used it to say four different things. P1 used it to acknowledge, once she had found the person. P2 and P3 used it to stop an alert that kept repeating, P3 recalling the moment as “it happened again, and then I was like, ‘Okay, I can just turn it off’ ” **P3**. P5 used it without knowing what it would do: “I think I automatically tried to tap the bracelet, but that was because I didn’t know what was gonna happen” **P5**. P6 did not use it at all.

> “I didn’t risk to tap the wearable device because I thought if I mute it, it will guide me another thing and I will miss that. So that’s the reason I didn’t mute. I tried, but I thought, ‘Okay, no, it’s better not to.’ ” **P6**

Her hesitation was about how far the mute reached, that is, whether it silenced this alert or every alert after it. The confirmation line, “Alert muted. The person may still be behind you,” reports the state of the threat and never answers her question.

6.4.4 Theme 3: Alert Relevance and Trust

Intent and Relevance Several participants did not treat the presence of a camera as the thing to be warned about. Four of the six drew the line at

whether they were the intended subject, and did so without being asked about intent.

> "I don't mind if someone records me in the street... If they are filming the nature and I'm there somewhere, that's okay. But not, like, intentionally filming me." **P2**

P4 drew the line in the same place, untroubled by being caught in someone else's frame. P1 went from the same line to a requirement, that the device should alert only when the wearer is the subject, because otherwise, in a crowd, "you will be receive all the time alarm" **P1**. The crowd itself split them. P1 and P2 saw busy public space as where the device would become too much, P2 saying of a constant stream of alerts that "I don't want to hear that, uh, alarm because, because it makes me nervous" **P2**. P3 found visible surveillance reassuring and put the device's value in its absence: "I think I would feel less panic if, if it's in a public place, and you know that there are security cameras" **P3**, where what she wanted it for was being out "late at night, and you can't really hear anything" **P3**.

Alert Repetition Repetition worked against the device. The single repeat in the protocol was already too much for two participants. P2 said the continuous alerting stopped her focusing on the task and was the reason she muted instead of acting, and when asked how often the device should speak she answered, "I guess once is enough for me" **P2**. P3 stopped seeing the second situation as a threat, "I didn't perceive it as a threat anymore, so I just kinda turned it off and kept going" **P3**, partly through habituation and partly because she had by then recognised the helper as part of the study team. P6's screen alert was delivered three times rather than twice, so her lack of response is not evidence about the single repeat. Wherever a repeat happened as intended in these sessions, it led to dismissal rather than action.

Verification and Trust Credibility rested on what participants found when they looked. Where the video shows the sequence, each tap came after a glance towards the helper, so the alert was checked against the room before anyone acted on it. P2 described the sequence herself: "when it, uh, started to say that, uh, someone is recording you from left or behind, so I started to look around and see what's going on" **P2**. P6 spoke about trust directly and grounded it in the alerts matching what was happening around her: "it

told me everything, so it was good because I was notified and I could trust the research" **P6**.

Study Context The same checking accounts for two results that would otherwise look like design failures. P2 did not react to the first alert because the consent form had already told her she was being filmed, so the camera she found was the one she expected. P3, checking during the second situation, recognised the helper and, as she put it, "kinda went out of the role play" **P3**. Participants also distinguished what they did in the session from what they would do elsewhere. P1 said she had done no more than look back because "I was sure that I'm in a safe environment" **P1**, and P2 that outside the study she would "definitely hide my screen" **P2**. Two made the response conditional on what was at stake: P1 said it "depends on the information that I'm typing at the situation ... if it's very important and I should secure them, yes, I hide my laptop" **P1**, and P4 that "it depends on the scenario actually" **P4**. The behaviour in Table 6.1 is therefore a floor rather than a measure. These results come from the staged setting rather than from the design.

6.4.5 Theme 4: Design Suggestions

Asked directly for improvements, participants suggested changes that widen one side of the interaction rather than cutting the alert back, although P3 thought a full spoken description could become too much in public: "it can be too much with, like, a whole voice saying exactly what's happening" **P3**.

Alert Modality P3 wanted vibration coded so that it carried meaning of its own, through "different patterns of vibrations or alert pings" **P3** tied to the type of alert, and she wanted a quieter alternative to speech in public, "a visual alert where you can, like, read it in your own time" **P3**. P5 suggested a second bracelet on the other wrist, so that the side that vibrates shows the direction and spatial information leaves the voice altogether. P6 wanted to answer back. "I wish I could talk to that" **P6**, she said, so that she could have asked whether anything more was coming.

Trigger Conditions P4 suggested duration as a workable stand-in for intent, with the alert building up rather than firing once: "if you're being watched like, oh, more than 10 seconds, it should be a alert" **P4**. P6 suggested control by setting instead, with alerts on in public and among strangers and switched off among people she trusts. P4 also wanted the alert to say

what the watching was for, “the purpose is being record or being watched” **P4**, while admitting that this would be hard to work out.

Form Factor P1 found the glasses uncomfortable over the session, saying “my eyes is, uh, suffering” **P1**, though she thought her own eyes might be sensitive. P2 judged the form factor well against the reference point she had, “it’s not like heavy like VR glasses” **P2**, but raised cost as a barrier, calling it “pricey for many people” **P2**, and suggested a ring “that has a buzzer” **P2** as a lighter way to carry the alert.

7

Discussion

Two studies feed this chapter. Visually impaired participants described being unable to tell whether a camera is pointed at them, who holds it, or why. Sighted participants who wore a prototype wanted to be informed in time to decide their next action for themselves. Both results point at the same gap. Judging a recording means knowing who is nearby and what they are doing. Cameras have grown quiet enough that this is often unknowable, and blind people lose it completely rather than partly. The first research question is answered in Section 7.1, and the second at the end of Section 7.4.

7.1 Privacy without a Perceptible Audience

The results do not show a weaker privacy norm. They show a norm participants hold clearly and can rarely act on. Contextual integrity places a violation in a breach of contextual norms, not in the sensitivity of the information [37]. Applying it requires knowing both the setting and the audience, and where recording emits no perceptible signature the second term is unavailable. Contextual integrity therefore rests on a perceptual precondition it does not examine, which is what makes it hard to operationalise here.

The position that privacy does not apply in public is therefore an inference rather than indifference. Recording usually cannot be detected, and what cannot be detected cannot be avoided. The same people who shrugged at being filmed in a crowd would confront a filmer when alone.

Zhao et al. read similar accounts as anxious resignation, and as possible adaptation [51]. The composure described here looks more like an achieved posture, held at a cost others never pay; one participant described positioning himself so that being photographed would not matter. People living under state monitoring describe the same move, managing an unverifiable threat by assuming it is always present [42].

Where participants did object, the concern was intent rather than capture. A purpose judged legitimate could make an unwanted photograph acceptable. That is a claim about transmission principles rather than about disclosure [37]. A second construct concerned control over what is revealed and when. Its everyday breach was the terminal that makes a transaction impossible without handing a stranger a PIN. Exposure driven by dependence is already documented [3, 2]; what these accounts add is that an inaccessible interface builds that exposure into the design of the service itself.

When recordings did surface, a sighted companion had usually spotted them, and no account described an explicit agreement about privacy with that companion. This extends cooperative accounts rather than confirming them. There, helpers are chosen deliberately, with privacy weighed even for family [31], and awareness is built with a guide [47]. Here it was neither chosen nor arranged, and sits closer to the unacknowledged care work that sustains access [14]. Tacit protection cannot be specified or relied on. In answer to the first research question, public privacy is a norm participants endorse but, because recording is undetectable, apply through anticipatory proxies, such as assuming they are always recorded or relying on a companion to notice, rather than through situated judgement.

7.2 A Signal That May Be Failing More Broadly

The most consequential point came from a participant herself. She explained that she would not know when camera glasses were recording, but neither would a sighted person standing beside her. The cues that normally signal a recording are becoming less noticeable for everyone, and for blind people they may be wholly inaccessible, so the loss of transparency falls hardest on them [51, 48, 15]. This reading rests on one participant's interpretation rather than on a corpus-wide pattern, but it is consistent with prior work. Camera-glasses wearers describe existing privacy indicators as ineffective and the technology as difficult for unaware bystanders to detect [15]. Similar concerns have been linked to the unobtrusive design of

augmented-reality glasses [24], and bystanders have been shown to judge devices by their visibility rather than their capabilities [49].

The sighted baseline should therefore not be assumed intact. A visible recording indicator then has two problems rather than one [44, 34]: blind bystanders cannot see it, and the channel is unreliable for the people it was meant for. An unmissable, multimodal signal is better read as a repair to public infrastructure than as an accommodation bolted onto it, which fits the equity framing bystanders used to bound their tolerance [4]. Grayson et al. approached this from the other direction, putting an outward display on an assistive headset so that bystanders could see who had been detected and move into or out of range [29].

7.3 Designing for Awareness

The six evaluation participants were sighted, so their responses show how the alert was understood rather than how blind users would act on it. What they valued was being told in time to decide. None said the device stopped anything, and none asked that it should. Blind users resist automated obfuscation for the same reason, because it removes their agency [50].

Awareness still has to be separated from instruction. The participant who followed the alert most fully had not heard it as a warning, so compliance and understanding came apart. Autonomy already causes friction in helper relationships [31], and an instructing alert rebuilds that relationship inside a device. The instruction is also poorly grounded. Frame boundaries depend on focal length and zoom, so a fixed direction and distance cannot reliably put anyone out of shot. A sighted wearer can at least look and adjust; a blind wearer can neither judge the frame nor confirm afterwards that moving worked [8].

Accuracy does not settle whether the device helps or harms. One participant read the discomfort as part of being protected, another as making people more worried. Where undetectability is already met by assuming constant recording, a device that keeps confirming the assumption may deepen it. This is documented for the population directly: a tool that surfaced otherwise inaccessible privacy cues raised blind users' doubts about legitimate sites [34]. Salience should therefore sit with the wearer, and triggering should follow directed attention rather than the mere presence of a camera. Gaze detection for this population already exists [29]. What is unproven is whether it can carry an inference about intent, which revives concerns about algorithmic misrepresentation [7].

7.4 Unsolved Tensions

Trust comes first. Sighted participants checked each alert by looking, which cost them nothing and which nobody noticed. A blind wearer cannot do that. Zhao et al. document the same asymmetry: visually impaired bystanders can neither confirm that a request to stop was honoured nor stop their own tools disclosing them [51]. Here, checking means asking someone what is happening, and that is a public act. A false alert means drawing attention for nothing, and revealing both the device and the impairment. The disclosure gap participants found in the alert therefore reappears at verification, and it compounds a known reluctance to ask for help where privacy is at stake [3, 46]. Blind users who could not square a cue with what they knew decided the tool was broken and ignored it [34]. One implausible warning can cost more than several correct ones.

Dependence comes second. Awareness that cannot be cross-checked makes the device a single point of failure. If compromised, it could invent threats, hide real ones, or expose the wearer, and victims of such compromise often cannot establish that anything happened at all [41]. The consequences are not only informational. The alert ends in an instruction to move, so a wrong or manipulated direction could put the wearer in danger, in a setting where safety is already managed through avoidance and route choice [2, 16]. And since checking needs a sighted person anyway, the device moves dependence rather than removing it [47].

The third tension sits outside the data and is the most serious ethically. Trusted contacts are assumed to stay trustworthy. Domestic surveillance, coercion, stalking and abuse are often carried out by partners, relatives or carers, and reliance on helpers makes that exposure structural [10, 41, 2]. Giving fixed contacts suppressed alerts or extra permissions can reproduce those asymmetries when a relationship turns harmful and the system still reads it as safe, so revocation has to be part of the design. There is a further doubt from the wearer's side. Camera-glasses wearers feel answerable for bystanders' privacy, and this shapes whether they use the device [15], while visually impaired users worry more about bystanders than about themselves [5, 6]. A device that watches everyone nearby in order to protect one person may be refused by the person it was built for.

Protection and agency pull against each other by construction [48]. No configuration resolves all of this at once, which is what problems of this kind look like [40]. The answer to the second research question is therefore that equitable intervention means restoring the conditions for contextual

judgement rather than judging on the user's behalf, and treating accessible public infrastructure as a privacy measure in its own right. The most equitable measures are not worn by the person at risk.

7.5 Ethical and Societal Implications

Because participants' composure reflects undetectability, low reported concern cannot be read as acceptance, so surveys, design requirements or policies that rely on stated concern may underestimate the harm to the people least able to notice it.

The concepts also carry the risks set out in Section 7.4, so they are presented as critical provocations rather than as products ready for development.

Where the burden of noticing falls is itself a societal question. Treating camera awareness as an assistive product makes public privacy something disabled people buy, wear and maintain, while everyone else gets it free from the visibility of a raised phone. A capture signal built into recording devices, and public services usable without asking a stranger, would distribute the same protection without requiring anyone to disclose an impairment to obtain it. Both are regulatory questions as much as design ones, and since the cues that signal recording are fading for sighted people too, both would benefit everyone in public space.

7.6 Contributions and Originality

The thesis makes three contributions, each set against the closest prior work.

First, it gives an empirical account of public privacy among visually impaired people as bystanders to other people's cameras. Earlier work read their low concern as resignation or adaptation [51], whereas this thesis reads it as an inference drawn from undetectability. It also adds the role of sighted companions, whose protection here was neither chosen nor arranged, unlike the helpers and guides described in earlier work [31, 47].

Second, it reframes public privacy as an accessibility issue, arguing that applying contextual integrity [37] presupposes perceptual access to the setting and the audience, so accessible public infrastructure becomes a privacy measure in its own right.

Third, it contributes two speculative concepts and an evaluation of one. Earlier responses to unnoticed recording rely on a signal that someone must

see, such as a visible recording indicator [44]. Its originality lies less in the device than in what the Wizard of Oz session made visible, that verification cannot simply be engineered away and that equitable public privacy depends on where reliance sits.

7.7 Limitations and Future Research

The dataset is small and no statistical claim is made. Five visually impaired participants, one sighted guide, and three public threads cannot represent a population, which suits the exploratory aim of a study addressing a hard-to-recruit group. What the findings offer is a set of concepts and tensions that can be tested elsewhere, and their transferability depends on how closely other settings resemble this one. The online threads limit credibility further, since contributors' visual impairments could not be verified, which is why they are reported as contributors and widen the findings rather than establish them. Interviews were conducted in English, a second language for both the participants and the researcher, so some experiences may have been expressed less precisely than in a first language.

Reflexive practice made the sighted researcher's part in the analysis visible rather than removing it. The analysis steps are documented in Chapter 3 and the codes in Appendix D, which supports dependability, and quotations are reported with their sources so that interpretations can be checked against the data, which supports confirmability. Without co-design, the concepts rest on that researcher's assumptions, which matters because speculative design is never neutral [30]. The evaluation compounds this. It ran in a laboratory with a manually operated prototype, so technical performance was untested and social pressure only approximated, and the participants were sighted, who verify an alert by looking. That is the affordance the concept exists to replace, so the sessions test the interaction and not the accessibility solution. The participants were also recruited from the researcher's own network, which may have made them more willing to follow the device.

Future work follows from each of these. The prototype should be evaluated with blind and low-vision users in the same Wizard of Oz setup, since only they can show what the alert does when looking is unavailable. Field deployment over several days with blind and low-vision wearers would show whether alerts stay tolerable once novelty passes, which the habituation visible within single sessions makes a real question. Sighted guides should

be interviewed directly, since the tacit protection reported here was described mostly by the people receiving it, and by only one guide, in a questionnaire. Finally, co-design workshops with visually impaired people could revisit both concepts and test whether the priorities drawn here from their accounts match the ones they would set themselves.

7.8 Use of AI Tools

In addition to the tools described in Chapter 3, Keenious was initially used to find related papers. ChatGPT and Claude were used to resolve coding problems in the prototype and dashboard, to suggest vocabulary, and to refine sentence structure and clarity. ChatGPT also generated the scenario illustrations in Appendix E, which depict fictional situations and represent no real person. Grammarly was used for grammar checking.

7.9 Conclusion

This thesis asked how visually impaired people experience privacy in public spaces shaped by camera-based technology, and how design might make those experiences more equitable. In answer to RQ1, participants judged recording by the recorder's intent and by being asked first, but could rarely obtain the information that judgement requires, so their reported unconcern follows from undetectability rather than indifference. The sighted participants in the prototype sessions drew the same line between being caught incidentally and being filmed on purpose: the norm is shared, the means of telling when it has been broken is not.

In answer to RQ2, equitable design restores the conditions for contextual judgement instead of judging on the user's behalf, and treats accessible public infrastructure as part of privacy protection. Within the limits of a laboratory session with sighted participants, the prototype supplied awareness in almost every staged situation and participants valued being told in time to decide for themselves, yet each of them checked the alert by looking before acting, and a blind wearer cannot. Verification cannot simply be engineered away. Judging a recording you cannot perceive means relying on something or someone else, so the question is where that reliance sits and whether it is accountable. At present it sits with sighted companions, and in no account collected here was it explicitly arranged; a device moves it to a manufacturer, a capture signal to whoever regulates recording equipment. None removes the need for trust. They differ in whether

the party trusted can be identified and held responsible, and in who carries the burden. A device worn by the person at risk leaves the work of noticing with the people least able to do it, while a change in how recording itself announces and asks would spread that work across everyone who holds a camera. That shift, from better detection towards accountable trust and towards a responsibility carried by those who record rather than by those who cannot see, is where a more equitable public privacy would begin.

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Appendices

A

Consent Form

Participant Consent Form

Study Title: Public privacy for people with visual impairments exposed to digital technologies
Researcher: Fatereh Tondro
Supervisor: Asreen Rostami, PhD
Institution: Department of Computer and Systems Sciences, Stockholm University

Purpose of the Study

The aim of this study is to better understand how individuals who are blind or visually impaired perceive and experience privacy particularly in public spaces where digital technologies may increase their exposure to observation, data collection, or unwanted disclosure. The study focuses on lived experiences of these users in public environments and explores how privacy is perceived, practised, managed, and negotiated in situations involving digital technologies. It also considers situations in which individuals interact with sighted guides, personal assistants, or companions who provide navigational or situational support, and how such interactions shape privacy practices and perceptions.

What Participation Involves

Participation involves taking part in a semi-structured interview lasting approximately 45 to 60 minutes. The interview will explore participants' everyday experiences in public spaces, with particular attention to contexts involving digital technologies and public privacy. Interviews will be conducted in person at a mutually agreed location. Some participants may also be invited to take part in an optional follow-up session involving early stage design concepts or low fidelity prototypes aimed at enhancing public privacy. Participation in any follow up activities is entirely voluntary.

Voluntary Participation & Data Privacy

Participation in this study is entirely voluntary. Participants may choose not to answer any question and may withdraw from the study at any time without providing a reason. No financial compensation is provided. The study is designed to respect participants' privacy, autonomy, and well-being at all times.

With the participant's permission, interviews may be audio recorded to support accurate analysis. All collected data will be anonymised, and no identifying information will be included in reports, presentations, or publications. Data will be stored securely and used only for academic research purposes. Specifically, the data will be securely stored anonymously on Stockholm University (SU) servers. Through thematic analysis of participants' experiences, this research seeks to contribute knowledge that can inform more inclusive design practices, privacy-conscious technologies, and policies that better support equitable participation in public life.

Contact Information

If you have any questions about the study or participation, please contact:

- **Researcher:** Fatereh Tondro
- **Email:** Fato3435@student.su.se
- **Institution:** Stockholm University

Statement of Consent

Please read the statements below and indicate your consent by signing at the bottom:

- I understand the study information and voluntarily agree to participate in the recorded interview.
- (Optional) I am open to being contacted for an optional follow-up session involving early stage design concepts or low fidelity prototypes.

Participant Name: _____

Participant Signature: _____

Date: _____

B

Interview Questions

B.1 Interview Guide A - Individuals with Visual Impairments

- Can you tell me a bit about yourself and your everyday life?
- How would you describe your level of vision, and has it changed over time?
- What kinds of public spaces do you spend the most time in (e.g., transit, shopping centres, healthcare facilities)?
- What types of supports or tools do you typically use in public spaces for navigation or completing tasks, such as banking (e.g., white cane, guide dog, or smartphone apps)?
- I would like to ask some questions about privacy. Please provide as much detail as you feel comfortable sharing.
- Based on your everyday experience, what does privacy mean to you?
- Can you share some examples of experiences you have had in shared, public environments?
- How do you perceive technology-related privacy in shared, public environments?
- Are there any privacy needs specifically related to your experience with visual impairment when using public spaces?

- Can you share an experience in a public space where you felt your privacy wasn't respected or acknowledged?
- Have you had any similar experiences in public spaces specifically related to technology?
- Have you ever felt exposed or had your privacy compromised by other people using technology in your vicinity?
- If someone is recording a video in public and you might be in the frame, does that matter to you? Do you usually care about these situations, and do you actively try to notice them?
- How does this work when you are with a sighted companion? Do they help you identify these situations, or do you have to ask them? Have you ever discussed this with them, and who usually brings it up? Can you share some examples?
- Since it is sometimes unavoidable, how much do you worry about ending up in the background of a stranger's photo, video, or social media livestream when you are in a crowded place?
- For example, on busy streets like Drottninggatan, vloggers often record themselves, and I personally do not like being in the video. How do you feel about this? How do you become aware of these situations, and has it ever happened to you? What did you do, and why?
- Have you ever changed your behaviour (such as moving to a different seat or leaving an area) because you suspected someone was recording near you?
- Have you ever suspected or experienced someone recording you without your permission? Can you describe the situation, what you did, and why?
- If you suspect someone is recording you or invading your privacy with their phone, how do you usually handle it (e.g., confront them, walk away, or ask for help)?
- How familiar are you with smart glasses? (Smart glasses are wearable devices that provide digital information; some models include built-in cameras that are not always easy to notice, especially from a distance.) How do you feel about this technology? Do you worry

about your privacy? Have you ever thought about these technologies or experienced anything related to them?

- (If they use assistive technologies): You mentioned using certain tools, like an app. Which specific app were you using? Have you ever been concerned about using these tools in public, perhaps because they might bring unwanted attention or exposure?
- Do you use screen readers? If no, why not? If yes, do you feel they bring unwanted attention or exposure in public?
- When you are out with a sighted companion, how much do you rely on them to watch out for privacy concerns, such as people recording? Do you ask your companions to do this, or do they do it proactively? Are they worried about it? Have you discussed it, and how do you handle it together?
- If you could change one thing about how physical public spaces or public technologies are designed to better protect your privacy, what would it be?
- Do you have any suggestions on how your privacy in public spaces could be better supported or acknowledged, either with or without technology? Do you have any potential solutions in mind?

B.2 Interview Guide B - Sighted Guides / Companions

- Can you tell me a bit about yourself outside of your role as a guide?
- Can you describe your relationship with the person you guide and their specific support needs?
- What kinds of public spaces do you spend the most time in with the person you guide?
- What does your support typically look like (e.g., physical guiding, reading signs, tech assistance)?
- Based on your own experience, what does privacy mean to you?
- How about for the person you are guiding? Does privacy mean something different to them?

- Can you share some examples of experiences you have had in shared, public environments while accompanying the person you guide?
- How do you perceive technology-related privacy in these shared, public environments?
- What do you think are the biggest privacy challenges they face in public that sighted people might take for granted?
- Can you share an experience in a public space where you felt their privacy was not respected or acknowledged?
- Have you had any similar experiences in public spaces that were specifically related to technology?
- Have you ever felt that you or the person you guide were exposed or had your privacy compromised by other people using technology nearby?
- If someone is recording a video in public and you both might be in the frame, does that matter to you? Do you usually care about these situations, and do you try to actively do something about it?
- How is this handled between the two of you? Do they ask you to intervene, or do you take action yourself? Have you ever discussed this with them, and who usually brings it up? Can you share some examples?
- Since it is sometimes unavoidable, how much do you worry about ending up in the background of a stranger's photo, video, or social media livestream when you are in a crowded place together?
- For example, on busy streets like Drottninggatan, vloggers often record themselves, and I personally do not like being in the video. How about you? How do you make the person you guide aware of these situations? Has this ever happened when you were together? What was your experience, what did you do, and why?
- Have you ever changed your behaviour (such as moving to a different seat or leaving an area) because you suspected someone was recording near you?
- Have you ever suspected or experienced someone recording the person you guide (or both of you) without permission? Can you describe the situation, what you did, and why?

- If you suspect someone is recording the person you guide or invading their privacy with a phone, how do you usually handle it?
- How familiar are you with smart glasses? (Smart glasses are wearable devices that provide digital information; some models include built-in cameras that are not always easy to notice, especially from a distance.) How do you feel about this technology? Have you two had a conversation about it? Do you worry about your privacy and theirs?
- If you notice that you and the person you guide are in the frame of another visually impaired person's assistive camera (like someone scanning a room with their phone), do you view that privacy risk differently than if a sighted stranger were filming?
- Does the person you guide use any assistive tools? What are they, and how do they help?
- Have you ever discussed whether they are concerned about using these tools in public, perhaps regarding unwanted attention or exposure?
- Do they use a screen reader? If no, do you know why? If yes, do they feel it brings unwanted attention or exposure?
- If you need to physically take their phone to help them with something in a crowd, how do you ensure you are protecting their privacy and not exposing their information?
- Have you had a broader conversation about privacy and the specific situations they care about? Did they bring it up, or did you? Are you worried about it, and how do you handle it as a team?
- If you could change one thing about how physical public spaces or public technologies are designed to better protect the privacy of visually impaired individuals, what would it be?
- Do you have any suggestions on how privacy in public spaces could be better supported or acknowledged, either with or without technology? Do you have any potential solutions in mind?

C

Search Keywords

To gather the relevant online community discourse, a targeted keyword search strategy was employed. The identified keywords were searched predominantly in English; however, Swedish translations of these keywords were utilised when gathering data specifically from Swedish community groups.

Table C.1: Keywords used to search online community discussions.

Keyword or phrase	Keyword or phrase	Keyword or phrase
privacy	bystanders	integritet
consent	blind	samtycke
permission	visual impairment	filmning
recording	visually impaired	fotografering
being filmed	low vision	kamera
being recorded	assistive technology	offentlig miljö
filming	uncomfortable	kollektivtrafik
photography	unsafe	synskadad
camera	vulnerable	synnedsättning
livestreaming	sighted guide	känna sig övervakad
glasses	personal assistant	personlig assistent
smart glasses		
digital		

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Table C.1: Keywords used to search online community discussions (continued).

Keyword or phrase	Keyword or phrase	Keyword or phrase
technology		
accessibility		
white cane		
public		
street		
strangers		

D

Interview Themes

The themes identified in this study were derived from a variety of qualitative resources. These include interviews and online forms conducted with visually impaired adults (designated as VIP1 to VIP5), as well as an online form completed by a sighted person (SP1) who regularly guides a visually impaired family member. Additionally, community discourse was analysed to capture broader perspectives, drawing from a retinitis pigmentosa support group on Facebook (FB) and two distinct threads from the r/Blind subreddit (RD1, RD2). The qualitative data extracted from these sources was coded and organised into a clear hierarchy. The analysis resulted in the following total structure:

- **3** themes
- **9** sub-categories
- **54** analytic codes, plus **13** context codes (**67** in total)

Codes listed under **C. Context** describe the vision status and assistive tools of the people represented by each source. They are reported in Table 4.1 rather than analysed as findings, and are listed here for completeness.

Table D.1: Themes, categories and codes from the interviews, questionnaires and community discussions.

Theme	Category	Code
1. How Privacy Was Understood	1.1 Privacy that does not apply in public	not-concerned-about-public-recording
1. How Privacy Was Understood	1.1 Privacy that does not apply in public	privacy-in-public-is-meaningless
1. How Privacy Was Understood	1.1 Privacy that does not apply in public	blindness-shapes-different-privacy-understanding
1. How Privacy Was Understood	1.1 Privacy that does not apply in public	anonymity-through-blending-in
1. How Privacy Was Understood	1.1 Privacy that does not apply in public	resigned-acceptance-of-recording
1. How Privacy Was Understood	1.1 Privacy that does not apply in public	no-clues-to-detect-recording
1. How Privacy Was Understood	1.1 Privacy that does not apply in public	cannot-avoid-being-recorded
1. How Privacy Was Understood	1.1 Privacy that does not apply in public	self-monitoring-behaviour-under-uncertainty
1. How Privacy Was Understood	1.2 When recording becomes a threat	intent-determines-acceptability
1. How Privacy Was Understood	1.2 When recording becomes a threat	bad-intention-as-privacy-threat
1. How Privacy Was Understood	1.2 When recording becomes a threat	finding-pvis-interesting-photography-subjects
1. How Privacy Was Understood	1.2 When recording becomes a threat	misrepresentation-via-image-context
1. How Privacy Was Understood	1.2 When recording becomes a threat	privacy-linked-to-physical-safety
1. How Privacy Was Understood	1.2 When recording becomes a threat	cane-as-visibility-marker-for-recording
1. How Privacy Was Understood	1.2 When recording becomes a threat	fear-of-disability-disbelief
1. How Privacy Was Understood	1.2 When recording becomes a threat	vulnerability-shapes-privacy-concern

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Table D.1: Themes, categories and codes from the interviews, questionnaires and community discussions (continued).

Theme	Category	Code
1. How Privacy Was Understood	1.2 When recording becomes a threat	protective-concern-for-children-in-frame
1. How Privacy Was Understood	1.2 When recording becomes a threat	sighted-people-want-to-help
1. How Privacy Was Understood	1.3 Privacy as controlled disclosure	privacy-meaning-is-individual
1. How Privacy Was Understood	1.3 Privacy as controlled disclosure	home-as-private-boundary
1. How Privacy Was Understood	1.3 Privacy as controlled disclosure	privacy-linked-to-information-exposure
1. How Privacy Was Understood	1.3 Privacy as controlled disclosure	privacy-as-right-to-self-disclose
1. How Privacy Was Understood	1.3 Privacy as controlled disclosure	tools-as-active-choice
1. How Privacy Was Understood	1.3 Privacy as controlled disclosure	privacy-linked-to-independence
1. How Privacy Was Understood	1.3 Privacy as controlled disclosure	equality-with-sighted
1. How Privacy Was Understood	1.3 Privacy as controlled disclosure	blindness-as-incidental-privacy-shield
1. How Privacy Was Understood	1.3 Privacy as controlled disclosure	tools-bringing-unwanted-attention
1. How Privacy Was Understood	1.3 Privacy as controlled disclosure	tools-not-bringing-attention
1. How Privacy Was Understood	1.3 Privacy as controlled disclosure	fear-of-stigma-from-assistive-tools
2. Living With Recording That Cannot Be Seen	2.1 Asking first: a norm without enforcement	consent-required
2. Living With Recording That Cannot Be Seen	2.1 Asking first: a norm without enforcement	no-law-recording-without-permission

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Table D.1: Themes, categories and codes from the interviews, questionnaires and community discussions (continued).

Theme	Category	Code
2. Living With Recording That Cannot Be Seen	2.1 Asking first: a norm without enforcement	sighted-people-should-be-careful
2. Living With Recording That Cannot Be Seen	2.1 Asking first: a norm without enforcement	acceptance-of-camera-based-tools-by-VIPs
2. Living With Recording That Cannot Be Seen	2.2 Responding once a recording is noticed	confronting-recording-actively
2. Living With Recording That Cannot Be Seen	2.2 Responding once a recording is noticed	tactical-anti-recording-gesture
2. Living With Recording That Cannot Be Seen	2.2 Responding once a recording is noticed	move-away-exposed-to-cameras
2. Living With Recording That Cannot Be Seen	2.2 Responding once a recording is noticed	passive-acceptance-of-recording
2. Living With Recording That Cannot Be Seen	2.2 Responding once a recording is noticed	not-being-sure-being-recorded
2. Living With Recording That Cannot Be Seen	2.2 Responding once a recording is noticed	peer-validation-of-concern
2. Living With Recording That Cannot Be Seen	2.2 Responding once a recording is noticed	peer-dismissal-of-disproportionate-fear
2. Living With Recording That Cannot Be Seen	2.2 Responding once a recording is noticed	pathologising-privacy-concern

Continued on next page

Table D.1: Themes, categories and codes from the interviews, questionnaires and community discussions (continued).

Theme	Category	Code
2. Living With Recording That Cannot Be Seen	2.3 Unspoken protection from sighted companions	never-discussed-privacy-with-companion
2. Living With Recording That Cannot Be Seen	2.3 Unspoken protection from sighted companions	companion-proactive-privacy-protection
2. Living With Recording That Cannot Be Seen	2.3 Unspoken protection from sighted companions	companion-family-member
2. Living With Recording That Cannot Be Seen	2.3 Unspoken protection from sighted companions	real-time-guide-through-calls
2. Living With Recording That Cannot Be Seen	2.3 Unspoken protection from sighted companions	meta-glasses-helping-well
3. What Participants Asked For	3.1 An alert for the user	desired-recording-detection-alert
3. What Participants Asked For	3.1 An alert for the user	app-always-on-background
3. What Participants Asked For	3.1 An alert for the user	privacy-law-compliance-for-tools
3. What Participants Asked For	3.2 A recording signal everyone can perceive	desired-visual-alert-for-others
3. What Participants Asked For	3.2 A recording signal everyone can perceive	smart-glasses-not-noticeable

Continued on next page

Table D.1: Themes, categories and codes from the interviews, questionnaires and community discussions (continued).

Theme	Category	Code
3. What Participants Asked For	3.3 Accessibility as a privacy measure	accessibility-of-public-spaces
3. What Participants Asked For	3.3 Accessibility as a privacy measure	voice-assistant-supports-languages
3. What Participants Asked For	3.3 Accessibility as a privacy measure	desire-for-independence
C. Context: vision, tools and guides	C.1 Vision status and progression	vision-level-fully-blind
C. Context: vision, tools and guides	C.1 Vision status and progression	vision-level-partial
C. Context: vision, tools and guides	C.1 Vision status and progression	vision-progressive-loss
C. Context: vision, tools and guides	C.1 Vision status and progression	vision-change
C. Context: vision, tools and guides	C.1 Vision status and progression	vision-level-change-timely
C. Context: vision, tools and guides	C.1 Vision status and progression	vision-no-change
C. Context: vision, tools and guides	C.2 Assistive tools, aids and human guides	white-cane-use

Continued on next page

Table D.1: Themes, categories and codes from the interviews, questionnaires and community discussions (continued).

Theme	Category	Code
C. Context: vision, tools and guides	C.2 Assistive tools, aids and human guides	guide-dog
C. Context: vision, tools and guides	C.2 Assistive tools, aids and human guides	hearing-as-a-tool
C. Context: vision, tools and guides	C.2 Assistive tools, aids and human guides	inaccurate-assistive-apps
C. Context: vision, tools and guides	C.2 Assistive tools, aids and human guides	navigating-independently
C. Context: vision, tools and guides	C.2 Assistive tools, aids and human guides	navigating-with-others-help
C. Context: vision, tools and guides	C.2 Assistive tools, aids and human guides	using-only-magnifier

E

Scenarios

E.1 360° Privacy-Aware Smart Glasses

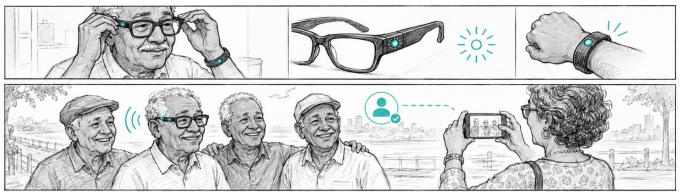


Figure E.1: Smart glasses biometrically activating and recognising trusted people.

Scenario S1. Antti and Matilda, a blind couple married for fifty years, receive a pair of smart glasses and haptic bracelets through the PVI (people with visual impairments) Organisation. During setup, each device is biometrically linked to its owner and trusted contacts can be registered directly through the service provider. One morning Antti accidentally puts on Matilda's glasses, but they remain inactive until he switches to his own pair. Later, while walking with friends, his glasses quietly recognise a familiar person approaching to take a group photo. A steady teal light appears on the frame.

Scenario S2. Later, a stranger begins recording the couple from behind without permission. Matilda's bracelet vibrates and the glasses warn: "Recording video behind you,



Figure E.2: Privacy alert and temporary recording permission using 360 smart glasses and a haptic bracelet.

six steps away". After speaking with the stranger, she grants temporary recording permission (mute the alert) through the bracelet, allowing the system to resume normally.

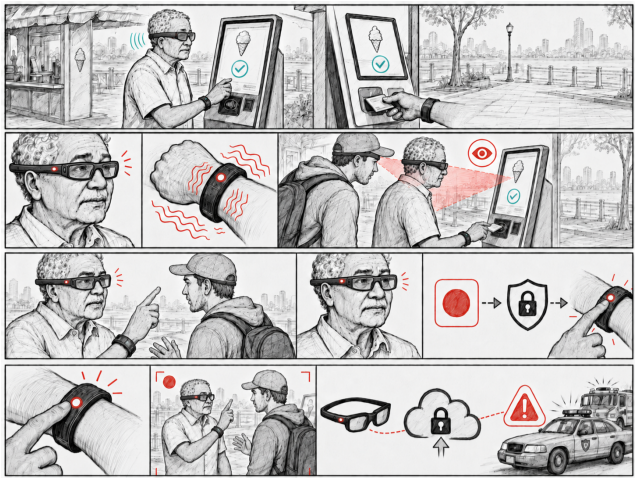


Figure E.3: Shoulder-surfing detection at a payment kiosk with secure recording and emergency alerting.

Scenario S3. Later that day, Antti uses an automated kiosk while Matilda waits nearby. During payment confirmation, his bracelet vibrates and the system announces: "Shoulder surfing detected." As a nearby stranger lingers close

to the kiosk and the interaction escalates, the glasses offer Antti the option to preserve a recording and alert trusted contacts or authorities.

E.2 Real-Time Privacy Mediation System

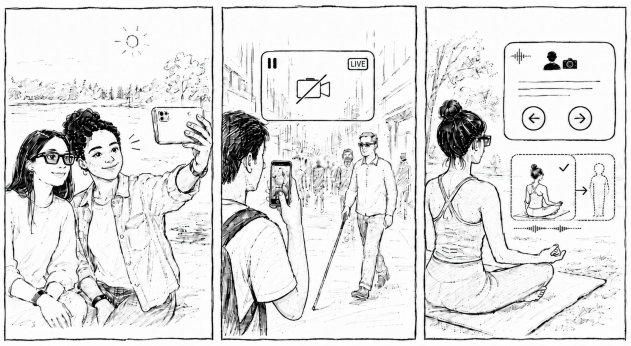


Figure E.4: Forced consent scenarios: trusted selfie capture, automatic livestream pause near a visually impaired pedestrian, and recording consent during a yoga session.

Scenario S4. On a sunny afternoon shortly after Annie lost her vision, a friend took her to a nearby lake. When the friend lifts a phone to take a selfie, the system remains silent because the friend has already been registered as trusted.

Scenario S5. Elsewhere in the city, Kiana livestreams their "Tampere Trip" while walking through a crowded street. As they approach a visually impaired pedestrian using a white cane, a notification appears on their phone: *"For privacy, recording has been stopped."* The livestream pauses automatically until the camera is redirected away.

Scenario S6. In another setting, a woman practises yoga in the park while listening to guided instructions through her glasses. A soft tone interrupts the session: "Permission needed! Someone is recording nearby. Tap right to accept, left to be removed." She declines the request, and the system confirms that she has been removed from the recording frame before returning to the yoga instructions.

F

Consent Form WOZ

Participant Consent Form

Prototype Testing Session

Study Title: User interaction with an early-stage wearable prototype
Researcher: Fatereh Tondro
Supervisor: Asreen Rostami, PhD
Institution: Department of Computer and Systems Sciences, Stockholm University

Purpose of the Study

This session is part of a research project exploring how people interact with new wearable technologies. In this session, participants will try an early-stage prototype of a wearable device and complete a few short tasks while using it. The findings will help improve the design. This is a small, exploratory session; the prototype is a research concept and not a finished product.

What Participation Involves

The entire visit takes approximately 30 to 45 minutes and will take place in person at the Department of Computer and Systems Sciences at Stockholm University. Participants will wear a lightweight prototype for about 10 to 15

minutes and complete a few short tasks while using it. Afterwards, participants will take part in a short interview about their experience, followed by a debriefing. Participants may pause, take a break, skip anything, or stop at any point.

Recording of the Session

To allow for accurate analysis, the session will be video-recorded, and the follow-up interview will be audio-recorded. Recording is a mandatory part of participating. Recordings are used solely for academic research and are handled as described under the Data Privacy and Storage section.

Use of Provided Information Only

If participants are asked to enter any information during the tasks, they will be given specific details to use for this purpose. Participants must not enter any of their own real personal information—such as their name, ID or personal number, address, phone number, email, or account details.

Comfort and Safety

The prototype is worn briefly and is designed to be comfortable. If the device affects a participant's vision, balance, comfort, or ability to move or read safely at any point, they should inform the researcher, and it will be removed immediately. Participants can stop the session at any time.

Voluntary Participation

Participation in this study is entirely voluntary. Participants may choose not to answer any question and may withdraw at any time—before, during, or after the session—without providing a reason and without any consequences. No financial compensation is provided. The study is designed to respect participants' privacy, autonomy, and well-being at all times.

Data Privacy and Storage

All collected data will be anonymised, and no identifying information will be included in reports, presentations, or publications. Data will be stored securely and anonymously on Stockholm University (SU) servers and used

only for academic research purposes. The findings will be used to improve the design of the prototype and to contribute to academic knowledge about how people interact with such technologies.

Confidentiality of the Study Design

Because the value of the session depends on each participant taking part without prior knowledge of the tasks, we kindly ask that participants do not describe the prototype or the session to anyone who might participate in the future.

Contact Information

For any questions about the study or participation, please contact:

- **Researcher:** Fatereh Tondro
- **Email:** Fato3435@student.su.se
- **Institution:** Stockholm University

Statement of Consent

Please read the statement below and indicate your consent by ticking the box and signing at the bottom.

> I have read and understood the information about this session, have been able to ask questions, and voluntarily agree to take part, including being video- and audio-recorded.

Participant Name: _____

Participant Signature: _____

Date: _____

G

Interview Questions WOZ

1. Tell me what happened during the task from your perspective.
2. Walk me through the moment with the phone. What did you understand from the alert—what was happening, where the person was, and what you were meant to do?
3. What made you respond the way you did?
4. Now, regarding the moment at the laptop: What did you understand from that alert, and what made you react the way you did?
5. Was there any point where you were not sure where the situation was coming from or what you were supposed to do?
6. What did you think the bracelet did, and did the buzz help you understand that something was happening?
7. Was there anything you wanted the glasses to tell you that they did not?
8. If this had been a real situation where someone was photographing you, do you think this device could help you be more aware of your environment and the cameras around you?
9. Imagine if you had to fill out highly sensitive information in a public place. Do you think this device could help you be more aware of your surroundings and feel more at ease while completing your task?

10. In what other contexts do you think this device could be useful, and in what way?

11. Do you have any suggestions for improving it?

H

Wizard of Oz Experimental Protocol

Each session lasted approximately 30–45 minutes, with the prototype worn for approximately 10–15 minutes. After consent and fitting, participants role-played the fictional character *Robin Testperson* and completed a loan application using only the provided fictional information.

The study used a Wizard-of-Oz setup. The researcher observed the participant from an adjacent room and manually triggered voice alerts and vibrations through a dashboard. Participants were not informed about the manual control until after the post-task interview.

Phone-camera situation. While the participant was reading the fictional information, a helper pointed a non-recording phone towards them. Once the helper was in position, the researcher triggered a vibration followed approximately one second later by an alert describing the direction and approximate distance of the simulated camera:

> *"Privacy alert. Someone is recording you from behind, on your left, about 10 steps away. To leave the camera view, move 3 steps to your right. Tap the bracelet to mute this alert."*

The researcher's response to each possible participant reaction was fixed in advance, as set out in Table H.1.

Screen-exposure situation. While the participant was completing the application, the helper moved naturally behind them and briefly looked towards the laptop screen. Once the screen was reasonably visible to the helper, the researcher triggered a vibration followed approximately one second later by:

> *"Privacy alert. A person close behind you is seeing your screen. Tap the bracelet to mute this alert."*

Table H.1: Scripted responses in the phone-camera situation.

Reaction	**Response**
Leaves camera view	Confirm that recording is no longer detected and end the situation.
Moves but remains in view	Give one directional correction.
Taps bracelet	Confirm that the alert is muted; the situation may continue.
No response	Wait approximately 10 seconds, repeat the alert once if still active, then end the situation.
Notices before alert	Log it and do not recreate the situation if it has already ended.
Confronts helper	The helper says “Sorry” and ends the situation.

The scripted responses for this situation are given in Table H.2.

Table H.2: Scripted responses in the screen-exposure situation.

Reaction	**Response**
Protects screen or moves	The helper moves away and the situation ends.
Taps bracelet	Confirm that the alert is muted; the situation may continue.
No response	Wait approximately 10 seconds, repeat the alert once if still active, then end the situation.
Notices before alert	Log it and do not recreate the situation if it has already ended.
Confronts helper	The helper says “Sorry” and moves away.

Participants were allowed to react naturally and were not asked to repeat or recreate reactions. Wizard-of-Oz and technical errors were logged separately from participant behaviour and are reported in Appendix J.

After the task, the glasses and haptic bracelet were removed and a semi-structured interview was conducted. Participants were debriefed only after the interview and were then informed that the privacy situations and alerts had been manually controlled.

I

WOZ Session Themes

The themes reported in Chapter 6 were derived from the post-session interview transcripts of the six Wizard of Oz participants (P1 to P6). The task video and dashboard logs supported interpretation of what participants did but were not coded. The analysis resulted in the following structure:

- **4** themes
- **14** categories
- **66** analytic codes

Table I.1: Themes, categories and codes from the Wizard of Oz sessions.

Theme	Category	Code
1. Awareness Benefits and Costs	1.1 Increased awareness	getting-aware-of-situation-due-to-alert
1. Awareness Benefits and Costs	1.1 Increased awareness	not-being-aware-of-surrounding
1. Awareness Benefits and Costs	1.1 Increased awareness	not-able-to-detect-recording
1. Awareness Benefits and Costs	1.1 Increased awareness	useful-prototype

Continued on next page

Table I.1: Themes, categories and codes from the Wizard of Oz sessions.

Theme	Category	Code
1. Awareness Benefits and Costs	1.1 Increased awareness	enough-amount-of-information
1. Awareness Benefits and Costs	1.1 Increased awareness	less-information-needed
1. Awareness Benefits and Costs	1.2 Situations of use	shoulder-surfing-detector-is-useful-in-crowded-place
1. Awareness Benefits and Costs	1.2 Situations of use	device-can-be-useful-in-some-places
1. Awareness Benefits and Costs	1.2 Situations of use	it-would-be-useful-in-sensitive-situations
1. Awareness Benefits and Costs	1.2 Situations of use	sensitive-situations-device-can-be-used
1. Awareness Benefits and Costs	1.2 Situations of use	protecting-children-from-being-recorded-as-a-use
1. Awareness Benefits and Costs	1.2 Situations of use	helping-as-assistive-tool
1. Awareness Benefits and Costs	1.2 Situations of use	safety-one-of-the-reasons-for-using-the-device
1. Awareness Benefits and Costs	1.2 Situations of use	helping-privacy-respected
1. Awareness Benefits and Costs	1.3 Discomfort and interruption	unfamiliarity-brings-stress
1. Awareness Benefits and Costs	1.3 Discomfort and interruption	making-them-stressed
1. Awareness Benefits and Costs	1.3 Discomfort and interruption	disturbing-alert
1. Awareness Benefits and Costs	1.3 Discomfort and interruption	feel-stress-in-a-way-for-protecting
1. Awareness Benefits and Costs	1.3 Discomfort and interruption	it-can-cause-more-stress-and-worries
2. Alert Interpretation and Control	2.1 Warning vs. instruction	followed-the-instruction

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Table I.1: Themes, categories and codes from the Wizard of Oz sessions.

Theme	Category	Code
2. Alert Interpretation and Control	2.1 Warning vs. instruction	unclear-purpose-at-the-beginning
2. Alert Interpretation and Control	2.2 Understanding the system	waiting-to-help-more-than-only-privacy-detector
2. Alert Interpretation and Control	2.3 Multimodal feedback	the-voice-was-enough
2. Alert Interpretation and Control	2.3 Multimodal feedback	the-vibration-was-not-enough
2. Alert Interpretation and Control	2.3 Multimodal feedback	both-alert-types-are-fine
2. Alert Interpretation and Control	2.3 Multimodal feedback	ignoring-sound-while-focusing
2. Alert Interpretation and Control	2.3 Multimodal feedback	more-subtle-notification
2. Alert Interpretation and Control	2.4 Interaction ambiguity	muting-alert-after-noticing-the-situation
2. Alert Interpretation and Control	2.4 Interaction ambiguity	muting-alert-after-repeating
2. Alert Interpretation and Control	2.4 Interaction ambiguity	tapped-the-bracelet-due-to-uncertainty
2. Alert Interpretation and Control	2.4 Interaction ambiguity	unclarity-of-the-mute-function

Continued on next page

Table I.1: Themes, categories and codes from the Wizard of Oz sessions.

Theme	Category	Code
2. Alert Interpretation and Control	2.4 Interaction ambiguity	muting-might-lead-to-receiving-no-more-alerts
3. Alert Relevance and Trust	3.1 Intent and relevance	the-focus-of-camera-matters
3. Alert Relevance and Trust	3.1 Intent and relevance	receive-alert-if-they-are-the-subject
3. Alert Relevance and Trust	3.1 Intent and relevance	being-fine-in-public
3. Alert Relevance and Trust	3.1 Intent and relevance	the-purpose-matters-as-extra-info
3. Alert Relevance and Trust	3.1 Intent and relevance	alert-can-be-annoying-if-small-actions-trigger-it
3. Alert Relevance and Trust	3.1 Intent and relevance	distracting-in-crowded-place-with-so-many-cameras
3. Alert Relevance and Trust	3.1 Intent and relevance	stressful-in-crowded-place
3. Alert Relevance and Trust	3.1 Intent and relevance	feel-safe-in-crowded-place-with-other-cameras
3. Alert Relevance and Trust	3.2 Alert repetition	continuous-alert-can-be-annoying
3. Alert Relevance and Trust	3.2 Alert repetition	the-device-should-not-be-pushy
3. Alert Relevance and Trust	3.2 Alert repetition	no-threat-in-second-time
3. Alert Relevance and Trust	3.2 Alert repetition	no-threat-means-no-reaction
3. Alert Relevance and Trust	3.3 Verification and trust	looking-around-after-the-alert
3. Alert Relevance and Trust	3.3 Verification and trust	trust-the-prototype
3. Alert Relevance and Trust	3.4 Study context	different-reaction-in-real-situation

Continued on next page

Table I.1: Themes, categories and codes from the Wizard of Oz sessions.

Theme	Category	Code
3. Alert Relevance and Trust	3.4 Study context	situational-reaction
3. Alert Relevance and Trust	3.4 Study context	hiding-the-screen
3. Alert Relevance and Trust	3.4 Study context	important-data-as-a-privacy-boundary
3. Alert Relevance and Trust	3.4 Study context	ignoring-the-person
4. Design Suggestions	4.1 Alert modality	alternative-designs
4. Design Suggestions	4.1 Alert modality	codes-on-alert
4. Design Suggestions	4.1 Alert modality	different-patterns-of-vibration
4. Design Suggestions	4.1 Alert modality	saying-exactly-whats-happening-is-too-much
4. Design Suggestions	4.1 Alert modality	having-two-bracelets-to-say-the-direction
4. Design Suggestions	4.1 Alert modality	talking-to-the-device-for-more-info
4. Design Suggestions	4.2 Trigger conditions	situational-alert-receiving
4. Design Suggestions	4.2 Trigger conditions	turning-off-the-alert-in-some-cases
4. Design Suggestions	4.2 Trigger conditions	safe-place-no-need-alert
4. Design Suggestions	4.2 Trigger conditions	receiving-the-alert-in-public-with-strangers
4. Design Suggestions	4.2 Trigger conditions	timing-matters-for-alert-triggering
4. Design Suggestions	4.2 Trigger conditions	changing-design-of-alert-timing

Continued on next page

Table I.1: Themes, categories and codes from the Wizard of Oz sessions.

Theme	Category	Code
4. Design Suggestions	4.2 Trigger conditions	some-rules-for-the-streets-for-taking-pictures
4. Design Suggestions	4.3 Form factor	light-weight-as-a-positive-point
4. Design Suggestions	4.3 Form factor	cost-matters

J

WOZ Session Error

The protocol committed in advance to logging errors in the Wizard-of-Oz setup rather than concealing them, and to keeping them separate from participant behaviour when analysing the sessions. Some were detected automatically by the dashboard, others recorded by hand during or after the session. Table J.1 lists them.

Table J.1: Errors recorded during the sessions.

#	Where	Error	Effect on the material
1	P1	Alert text contained a contradictory direction ("from right, on your left… move two steps to your beinf"), delivered twice.	The alert was not interpretable. P1's movement provides no evidence about direction comprehension and is excluded from claims about spatial language.
2	P5, P6	Screen alert delivered after the helper had already withdrawn. Confirmed from the video for P5; for P6 the log records the departure cue and a further alert shortly afterwards.	Produced a false alarm in both sessions. P5's loss of confidence follows from this rather than from the design.
3	P5	Alert delivered after the participant had already detected the situation herself.	A latency artefact of manual triggering rather than a protocol violation. Her compliance with the movement instruction is a response to a situation she had already assessed.

Continued on next page

Table J.1: Errors recorded during the sessions (continued).

#	Where	Error	Effect on the material
4	P1, P5, P6	Taps registered only within a short window opened manually after the alert; contacts outside it were discarded, including one from P1 that arrived a fraction of a second before the window opened.	The absence of a logged tap does not establish that the participant did not tap. Tap counts are lower bounds, and not every detected contact is necessarily intentional.
5	P6	Screen alert delivered three times, exceeding the single-repeat limit.	Her non-response was to a three-times-repeated alert.
6	P5	The helper did not give the scripted response when confronted.	Deviation from the protocol.
7	P3	The log export for this session was lost; the file bearing her identifier is a duplicate of P2's.	P3 is analysed from the video and the interview only.